ROOT CAUSE UNLOCKED · ULTIMATE GUIDE

Unlock Your *Fibromyalgia*, Ultimate Guide

Why the Pain Is Real, Why Your Labs Are Normal, and What Actually Moves the Needle

A guide for anyone who has been told their bloodwork looks fine, been handed a prescription and a shrug, and left the appointment with the same pain they walked in with.

Dr. Adam Seay, DC

Root Cause Unlocked, Garner, NC · rootcauseunlocked.com · (919) 662-0520

Unlock Your Fibromyalgia, Ultimate Guide

Why the Pain Is Real, Why Your Labs Are Normal, and What Actually Moves the Needle

> **BEFORE YOU START**

Contents

Before anything else: your pain is real, and I am going to be honest with you

Two things are true at once, and most books on this subject only tell you one of them.

Your pain is real. It is not imagined, not exaggerated, and not a personality trait. There is a measurable difference in how your nervous system processes pain signals, and it has a name.

And there is no cure. Anyone selling you one is lying to you, and the people with fibromyalgia I have met have already been lied to enough.

What sits between those two facts is a large amount of genuinely useful ground: what is actually happening, which of the things you have been told are wrong, what has real evidence behind it, and what a realistic good outcome looks like. That is what this book is for.

A note on what this book is not. It does not tell you what supplements to take, what to eat, or how to read your lab results. Those are the three companion guides, the diet guide, the supplement guide, and the lab guide, and they are built to be used rather than read. This book is the understanding. If you want the doing, that is where it lives.

Red flags that need medical attention, not a self-help plan

Fibromyalgia does not cause these. If you have any of them, they need evaluation on their own terms.

- Fever, night sweats, or unexplained weight loss
- Joint swelling, redness, or heat, as distinct from aching
- Sudden or severe weakness, particularly on one side
- New numbness with loss of bladder or bowel control
- A rash accompanying joint pain
- Pain that is truly new, truly localized, and steadily worsening

Widespread aching that migrates, fatigue that sleep does not fix, and pain that varies with weather, stress, and activity is the fibromyalgia pattern. The list above is not.

INTRODUCTION

The pain nobody can find

You know the appointment. You describe pain that is everywhere and nowhere in particular. You say you are exhausted in a way that sleeping does not touch. You mention the fog, the way words go missing mid-sentence. The physician runs a panel. The panel comes back clean.

And then something happens that does more damage than the pain: the clean panel gets treated as an answer instead of a finding. You get told there is nothing wrong. Some people get told it more gently than others. A great many get told, in one phrasing or another, that it might be stress.

Here is the thing that appointment got backwards. **Normal labs are not evidence against fibromyalgia. They are the expected result.** There is no blood test that confirms it. That is a known feature of the condition, not a hole in your story, and a clinician who understands fibromyalgia is not surprised by a clean panel. They are checking for something else entirely, which is the subject of one of the keys below.

Fibromyalgia is common. A 2026 systematic review of prevalence in the general population and at-risk groups notes that estimates vary widely worldwide, driven by differences in diagnostic criteria and study methods.[1] The variability itself tells you something: this is a condition defined clinically rather than by a test, and where you draw the line changes how many people fall inside it.

What follows are eight things worth understanding. Not a protocol. Understanding, which is the part that has been missing.

KEY 1

What is actually happening is a change in pain processing, not damage in the tissue

The single most useful idea in this book is that fibromyalgia is a problem with the volume knob, not the speakers.

In most pain, something is damaged and nerves report it. In fibromyalgia the reporting system itself has changed. Signals that should register as pressure register as pain. Signals that should fade keep firing. The medical term for this mechanism is **central sensitisation**, and a 2026 review of the year's research states plainly that it remains the main neurobiological mechanism, supported by the accumulated evidence.[2]

There is a broader category name worth knowing too, because you will encounter it: **nociplastic pain**, defined as altered nociceptive processing without clear tissue damage or a lesion in the sensory system. Fibromyalgia is the flagship example.

Why this matters practically, in three ways.

It explains why imaging and bloodwork are clean. They are looking at the speakers.

It explains why the pain moves, varies, and does not respect anatomical boundaries the way an injury does. A sensitized system responds to context: sleep, stress, weather, exertion.

And it explains why treatments aimed at tissue tend to disappoint, while treatments aimed at the nervous system tend to do better. That single insight redirects a lot of wasted effort.

What it does not mean. It does not mean the pain is psychological. Central sensitisation is a change in how a physical system processes signals. Saying pain is generated centrally is not the same as saying it is imagined, and anyone who slides from one to the other is either confused or being unkind.

KEY 2

Your labs are normal because that is what fibromyalgia looks like, and testing still matters

If there is no confirmatory test, why test at all?

Because the point of testing in fibromyalgia is not to find fibromyalgia. It is to find the conditions that look exactly like it and are treated completely differently.

Hypothyroidism produces fatigue, aching, and cognitive fog. So does low B12, and low B12 can also produce nerve symptoms that are permanent if left long enough. Iron deficiency produces exhaustion and poor exercise tolerance well before it produces anemia. Vitamin D deficiency produces diffuse musculoskeletal pain. Inflammatory arthritis in its early stages can present as widespread ache before joints visibly change. Sleep apnea produces unrefreshing sleep, fatigue, and pain amplification, and is routinely missed in women.

Every one of those is more treatable than fibromyalgia. Every one is a genuine miss if it is not looked for.

So the sequence that makes sense is: rule out the mimics, then treat what remains. A clean panel after a thorough look is a useful, meaningful result. A clean panel from a basic screen ordered to end the conversation is not the same thing, and it is worth knowing which one you got.

The companion lab guide covers exactly which panels answer which question, what the numbers mean, and includes a request list you can hand to a clinician. I am deliberately not reproducing it here.

One thing to be careful about. There is a claim circulating that fibromyalgia is caused by insulin resistance, based on a 2019 paper. **That paper was retracted.** It is not evidence, and it should not be the basis of anyone's treatment decisions. I mention it only because it still circulates widely and you deserve to know what you are looking at when you meet it.

KEY 3

Sleep and pain feed each other, and this loop is the most actionable thing in the book

Almost everyone with fibromyalgia sleeps badly. The usual assumption is that pain wrecks sleep. That is true, and it is only half the loop.

A study tracking daily sleep and pain ratings in chronic pain patients found a genuinely **bidirectional** relationship: a night of poor sleep was followed by increased pain the next day, and a day of increased pain was followed by a night of poor sleep.[3] Depression scores influenced both directions.

That bidirectionality is good news, oddly, because it means the loop has two entry points and one of them is more tractable than the other. You cannot decide to have less pain tomorrow. You can change several things about tonight.

Why sleep is where I would start with most people. It is measurable within days rather than months. It affects fatigue, fog, and mood as well as pain. And improvement compounds, because a better night lowers tomorrow's pain, which makes tomorrow night easier.

What this is not. It is not a claim that fixing sleep fixes fibromyalgia. It is a claim that of all the loops running, this is the one where effort pays back soonest, and that starting where the leverage is beats starting where the frustration is.

The practical specifics on eating for sleep are in the companion diet guide, and the supplement options with their evidence and interaction warnings are in the companion supplement guide.

KEY 4

Movement helps, and doing it wrong sets you back for weeks

This is the key where generic advice does the most harm, so I want to be precise.

Movement has real evidence. A 2026 systematic review and meta-analysis of 23 randomized trials covering 1,734 participants found that mindful movement approaches beat control conditions across measures including fatigue, with a standardized mean difference of 0.63.[4] Notably, sleep quality was one of the measures where the benefit did not hold, which is a useful reminder that no single intervention covers everything. A separate 2026 analysis of 21 trials with 1,638 participants compared aerobic exercise, resistance training, and other modalities directly.[5]

So the question is not whether to move. It is how.

The mistake almost everyone makes. You have a good day. You feel briefly like yourself. You do everything you have been putting off. And then you lose the next three days, or the next week, and conclude that exercise makes fibromyalgia worse.

It is not that exercise made it worse. It is that the dose was set by how you felt in the moment rather than by what you could reliably repeat.

Pace by what is repeatable, not by capacity. The practical version: find the amount you could do on a bad day, do that amount on every day including good ones, and increase it slowly when the baseline itself has moved. This feels absurdly conservative for the first few weeks. It is also the difference between a line that trends up and a sawtooth that goes nowhere for years.

A specific and important caution. If you experience post-exertional malaise, meaning a delayed and disproportionate crash in the day or two after activity, do not follow a graded exercise programme that requires steady weekly increases regardless of symptoms. That prescription has caused real harm in people with post-exertional malaise. Pacing is the approach that fits, and the distinction is not academic.

KEY 5

Stress is part of the mechanism, and that is not the same as it being your fault

This key gets skipped in a lot of patient material because it is easy to say badly. I would rather say it carefully than leave it out.

The nervous system that processes pain is the same one that runs your stress response. They are not separate systems that occasionally interact. A system held in a state of threat processes sensory input differently, and it does so through physical mechanisms, not attitude.

What follows from that, and what does not.

It follows that stress load genuinely affects symptom intensity, that this is worth working with, and that approaches aimed at the nervous system rather than the tissue have a legitimate place.

It does not follow that fibromyalgia is caused by stress, that you brought it on yourself, or that you could think your way out of it if you tried harder. That framing is both wrong and cruel, and if a clinician has offered it to you, they were wrong.

There is a related trap on the other side. Because stress is part of the mechanism, some people conclude the whole problem is psychological and go looking for a psychological cure. Psychological approaches help with the suffering and the disability, and they are worth doing. They do not switch off central sensitisation.

KEY 6

What the medications actually do, including the part your prescriber may not have mentioned

Three drugs are FDA-approved for fibromyalgia in the United States: pregabalin, duloxetine, and milnacipran. Here is what the evidence actually shows about them, which is more sobering than their approval status suggests.

The Cochrane review of SNRIs for fibromyalgia concluded, on low to very low quality evidence, that duloxetine and milnacipran provided **no clinically relevant benefit over placebo** in the frequency of achieving 50% or greater pain relief. They did show benefit on patients' global impression of being much or very much improved.[6] That is a meaningful distinction: people felt better overall more often than they achieved substantial pain reduction.

And a network meta-analysis of 36 studies covering 11,930 patients compared amitriptyline, an old and inexpensive generic used off-label, against the three FDA-approved options.[7] Compared with placebo, amitriptyline was associated with reduced sleep disturbance with a standardized mean difference of 0.97, and reduced fatigue with a standardized mean difference of 0.64. Those are respectable numbers for the drug that is not the approved one.

What I want you to take from this. Not that you should stop a medication, and not that you should start one. None of these are things I prescribe. They are included here so you know what exists and can raise it with the physician managing that part of your care. Both starting and stopping are decisions for you and your prescriber, and stopping a psychiatric or neurological medication abruptly on the strength of a book is a genuinely bad idea.

What I want you to take is that the approved-versus-off-label distinction reflects who ran trials for an indication, not a clean ranking of what works. If a medication is helping you, that is real regardless of what the pooled data says about averages. If one is not helping after a fair trial at an adequate dose, the fact that it is FDA-approved for your condition does not obligate you to keep taking it, and that is a conversation worth having rather than a verdict to accept.

KEY 7

The comorbidities are not coincidences

Fibromyalgia travels with a specific set of companions, and treating them as separate problems handled by separate specialists is how people end up with six providers and no plan.

The usual cluster includes irritable bowel syndrome, migraine or chronic headache, temporomandibular joint pain, restless legs, interstitial cystitis or bladder pain, anxiety and depression, and orthostatic symptoms such as dizziness on standing.

These are not unrelated bad luck. They share the sensitization mechanism. A nervous system amplifying signals from muscle also amplifies signals from gut, bladder, and blood vessels. Recognizing that reframes the situation usefully: you do not have seven conditions, you have one process expressing itself in seven places.

Practically, that means two things. An intervention aimed at the underlying process can improve several of them at once, which is why sleep and pacing show up repeatedly. And symptoms should still be evaluated individually rather than attributed to fibromyalgia by default, because people with fibromyalgia also get appendicitis and thyroid disease like everyone else. Attributing everything to the diagnosis is the mirror-image error of attributing nothing to it.

KEY 8

What realistic improvement actually looks like

I would rather give you an accurate picture than an inspiring one, because inaccurate expectations are how people conclude they have failed at something that was never on offer.

What is realistic. Meaningfully fewer bad days. Bad days that are less severe and shorter. More predictability, which matters more than most people expect, because unpredictability is what makes planning impossible and planning is what makes a life. Better function at the same pain level, which is a real win even though it does not look like one on a pain scale. Improvement measured over months and seasons rather than weeks.

What is not realistic. A cure. A pain-free baseline. A single intervention that resolves it. Linear progress without setbacks.

How to tell whether something is working, given all of that. Not by how you feel on any given day, which is noise. By whether the trend over a month has moved, by whether your bad days are less bad, and by whether you can do more before symptoms escalate.

This is precisely why tracking matters more in fibromyalgia than in most conditions. The changes are gradual, memory is unreliable about gradual change in both directions, and without a record you are left comparing today against an inaccurate memory of six months ago. That comparison usually understates real improvement, which then gets abandoned as useless.

A staged roadmap

Not a protocol, and not a schedule you are failing if you deviate from. An order of operations that puts effort where it pays back soonest.

First, establish the baseline and rule out the mimics. Get the testing done properly, with the request list from the companion lab guide if that helps. Start tracking before changing anything, so you have a real starting point rather than a remembered one.

Then work on sleep, because it has the shortest feedback loop and it lowers tomorrow's pain.

Then establish a repeatable movement floor, set by what you could do on a bad day, held steady for several weeks before it moves.

Then, one at a time, run the specific experiments in the companion diet and supplement guides. One at a time is not a stylistic preference. Running four changes at once produces a feeling instead of a finding.

Reassess at three months, not three weeks. Fibromyalgia moves on a slower clock than most conditions, and judging an intervention at two weeks will make you abandon things that would have worked.

Take the next step: your free Root Discovery Session

If you have read this far, you probably have a version of the same question: which of these applies to me, and in what order.

That is the conversation a book cannot have with you. It cannot see your labs, your medication list, which symptom is costing you most, or which of the mimics has actually been ruled out rather than assumed.

Book a free Root Discovery Session: rootcauseunlocked.com/discovery?utm_source=ebook&utm_medium=ebook&utm_campaign=fibromyalgia

There is no obligation to continue, and if your case belongs with someone else, I will tell you so.

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

Get in touch: rootcauseunlocked.com/contact?utm_source=ebook&utm_medium=ebook&utm_campaign=fibromyalgia

If you would rather start on your own, start tracking: rootcauseunlocked.com/tracker?utm_source=ebook&utm_medium=ebook&utm_campaign=fibromyalgia

The tracker is free, works anywhere, and your entries stay on your own device. Given how much of managing fibromyalgia depends on seeing a trend you cannot feel day to day, this is genuinely one of the higher-value things you can do this week, and it costs nothing.

Disclaimer

This book is educational. It does not constitute medical advice, diagnosis, or treatment, and reading it does not establish a patient-provider relationship. Nothing here is a reason to start, stop, or change a prescription on your own. Fibromyalgia shares symptoms with several conditions that are treated differently, which is why the testing chapter exists and why an accurate diagnosis belongs with a clinician who has examined you.

References

1. The Prevalence of Fibromyalgia in the General Population and At-Risk Subpopulations: A Systematic Review. *Anesth Analg.* 2026. PMID 42201834.
2. Pathogenetic and clinical aspects of fibromyalgia: one year in review 2026. *Clin Exp Rheumatol.* 2026. PMID 42328952.
3. Affleck G, et al. Intraindividual variability in daily sleep and pain ratings among chronic pain patients: bidirectional association and the role of negative mood. *Clin J Pain.* 2011. PMID 21415723.
4. Beyond pain: Impact of movement-based mindful exercises in fibromyalgia. A systematic review and meta-analysis. *J Bodyw Mov Ther.* 2026. PMID 42264784.
5. Comparison of the Effects of Three Types of Exercise on Fibromyalgia Syndrome. *Pain Res Manag.* 2026. PMID 41919680.
6. Welsch P, Üçeyler N, Klose P, Walitt B, Häuser W. Serotonin and noradrenaline reuptake inhibitors (SNRIs) for fibromyalgia. *Cochrane Database Syst Rev.* 2018;2:CD010292. PMID 29489029.
7. Farag HM, et al. Comparison of Amitriptyline and US Food and Drug Administration-Approved Treatments for Fibromyalgia: A Systematic Review and Network Meta-analysis. *JAMA Netw Open.* 2022;5(5):e2212939. PMID 35587348.

Root Health | Dr. Adam Seay, DC | Garner, NC | rootcauseunlocked.com

Working with Root Health

This guide is general education by design, written so it is useful without knowing anything about your labs. Personalized work is different: it starts with testing, your history, your medication list, and the symptoms costing you the most, then sequences changes instead of piling them on.

If you want that, a Root Discovery Session is free and is the right first step. Book at rootcauseunlocked.com/discovery.

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

If you would rather start on your own, the symptom tracker at rootcauseunlocked.com/tracker is free and works anywhere. In fibromyalgia it does real work, because the changes are gradual and memory is an unreliable instrument across weeks.

Reach me at (919) 662-0520 or at 503 US-70 East, Suite L, Garner, NC 27529.

From Understanding to Doing

Everything up to this point was about understanding: what is happening in your body, why it may be happening, and which directions are worth investigating. That is the half a book can do well.

What follows is the other half, and it is a different kind of document on purpose. Three guides, written to be used rather than read straight through. The first tells you which lab panels answer which question and what the numbers mean, and it includes a request list you can hand to a clinician. The second covers what to eat and, more usefully, how to test a dietary change on yourself so you end up with an answer instead of a habit. The third walks through what has actually been studied, at what dose, and names the products whose trials came back negative.

Each one stands alone, so use them in whatever order matches what you are stuck on. If you are not sure where to begin, begin with the lab guide, because knowing what is driving your symptoms changes what you should do about the other two.

A note on repetition. A few points appear in both halves. That is deliberate. The ones repeated are the ones where getting it wrong causes harm, and they are worth meeting twice.

Please Read This First

This guide is education, not medical care. It does not diagnose you, it does not treat you, and it does not replace your physician, rheumatologist, or any other member of your care team. Reading it does not create a patient-provider relationship with me or with Root Health.

No lab test in this guide can confirm or rule out fibromyalgia. That is not a limitation of this guide. It is a fact about the condition itself, and the rest of this document explains why that is actually useful information rather than bad news.

Never stop, start, or change the dose of any prescription medication because of something you read here. Some medications legitimately affect the lab values discussed in these pages. If that applies to you, the correct move is to tell your prescriber what you noticed, not to adjust anything yourself.

Every result in this guide has to be interpreted by a clinician who knows your full history. Numbers on a page do not carry meaning on their own. They carry meaning next to your symptoms, your exam, your medications, your family history, and your timeline. Bring your actual results to your doctor or to a Root Discovery Session.

If a result alarms you, or if you have any of the warning signs listed in the final section, get clinical review promptly rather than waiting.

The Honest Starting Point

Let us get the most important thing out of the way in the first paragraph, because most guides bury it.

There is no blood test for fibromyalgia. There is no antibody, no inflammatory marker, no hormone, no scan, and no genetic panel that confirms it. Fibromyalgia is diagnosed clinically, from your symptom pattern, using criteria built around widespread pain, fatigue, unrefreshing sleep, and cognitive symptoms, and it is confirmed by a clinician's judgment rather than by a laboratory (PMID 24737367). The 2016 revision of the diagnostic criteria is entirely symptom-based, built from a widespread pain index and a symptom severity scale, with no laboratory component at all (PMID 27916278).

You have probably been told some version of "your labs came back normal." If you took that as a verdict on whether your pain is real, please set that interpretation down. Normal labs are the expected finding in fibromyalgia. They are consistent with the diagnosis, not evidence against it.

So why run labs at all?

Because labs do three genuinely useful jobs, and none of them is "prove you have fibromyalgia."

Job one: rule out the conditions that look like fibromyalgia. Widespread pain and crushing fatigue are not unique to fibromyalgia. Thyroid disease, iron deficiency, vitamin D deficiency, B12 deficiency, celiac disease, inflammatory arthritis, and sleep apnea can all produce a picture that looks a great deal like it. Some of those are straightforward to correct. Missing one of them is the single most expensive mistake in this whole process.

Job two: find the things riding alongside it. Fibromyalgia is not exclusive. It coexists with other rheumatic diseases at meaningful rates, and someone with an inflammatory arthritis can also have fibromyalgia at the same time (PMID 24589726). One diagnosis does not close the file on the others.

Job three: describe the terrain. This is the part that most workups skip. Blood sugar regulation, inflammatory load, nutrient status, stress-axis function, and sleep quality all shape how much pain a nervous system produces and how much energy you have to spend on a given day. None of these is a fibromyalgia test. All of them are levers, and a lever you can see is a lever you can pull.

That is the honest value of testing here. Not a diagnosis. A map.

How to Read This Guide

Sections 1 and 2 cover the rule-out tier, the labs most physicians will recognize immediately and can order without any convincing. Section 3 covers the terrain tier, the labs that describe the environment your nervous system is operating in. Section 4 is about the difference between “in range” and “optimal,” which is where functional medicine and conventional medicine most often talk past each other, and where you deserve a straight answer about what is settled and what is a clinical preference. It includes the actual target windows I use at Root Health. Section 5 shows you how to read your results as a pattern. Section 6 gives you the actual conversation to have with your doctor, including a request list you can print. Section 7 covers what labs cannot tell you. Section 8 covers when to stop reading and go get seen.

You do not need every test in here. Section 6 tells you where to start.

Section 1: The Rule-Out Tier, Part One (Thyroid, Iron, Vitamin D, B12)

These four account for most of the correctable conditions that get mistaken for, or that quietly worsen, a fibromyalgia picture. If you only ever get one round of labs, this is the round.

Thyroid

Underactive thyroid produces fatigue, muscle aches, cold intolerance, cognitive slowing, and low mood. Read that list again and notice how much of it overlaps with what brought you here.

The relationship goes deeper than symptom overlap. In one small controlled study, roughly a third of patients with Hashimoto's thyroiditis, the autoimmune form of thyroid disease, also met criteria for fibromyalgia, and the association tracked with the presence of thyroid antibodies rather than with subclinical hypothyroidism alone (PMID 21085966). That is a small study and it should be read as a signal rather than a settled number, but it is the reason a thorough thyroid look matters here.

What most people get ordered: TSH alone.

What is worth asking about: TSH, free T4, free T3, thyroid peroxidase antibodies (TPO), and thyroglobulin antibodies (TgAb).

Why the antibodies specifically? Because thyroid autoimmunity can be present and active while TSH still sits inside the reference range. A TSH-only panel is designed to answer "is this gland currently failing," which is a good question. It is not designed to answer "is this gland under autoimmune attack," which is a different question and the one that maps onto the association described above.

How to read it honestly. A positive TPO antibody does not mean you have fibromyalgia and it does not mean you have hypothyroidism today. It means autoimmune thyroid activity is present, which is worth knowing, worth monitoring, and worth discussing with your physician. A normal TSH with positive antibodies is a real and common pattern that deserves follow-up rather than dismissal.

Iron and Ferritin

Iron deficiency produces fatigue, exercise intolerance, poor sleep, and restless legs, all of which are already common in fibromyalgia. It can be present without anemia, which means a normal hemoglobin on a standard complete blood count does not settle the question.

The evidence here is genuinely mixed, and you should know that before you go asking for it. One case-control study found significantly lower mean ferritin in fibromyalgia patients than in matched controls, and reported that a ferritin below 50 ng/mL was associated with increased odds of fibromyalgia even when the value sat inside the laboratory's normal range (PMID 20087382). A separate study of non-anemic fibromyalgia patients found no difference in iron, ferritin, transferrin, or soluble transferrin receptor compared with controls, and concluded there was no evidence to support iron supplementation as a fibromyalgia treatment (PMID 22095117).

A randomized placebo-controlled trial of intravenous iron in patients who had both fibromyalgia and documented iron deficiency did not meet its primary endpoint, though it reported greater improvement than placebo on several secondary measures (PMID 29149437). That is a legitimately encouraging signal and it is also, honestly, a trial that missed its main target. Both things are true.

Where does that leave you? Iron status is worth checking because iron deficiency is common, correctable, and produces symptoms you already have. It is not worth treating as a fibromyalgia cure. And it is worth knowing that the exact ferritin threshold that should count as deficiency is itself an unsettled question in the literature, not a fixed line (PMID 34028001).

What to ask for: complete blood count with differential, ferritin, serum iron, total iron binding capacity, and transferrin saturation. Ask for ferritin explicitly. It is frequently left off.

One important caution: ferritin is an acute phase reactant, which means inflammation, infection, and liver issues can push it up and mask a real deficiency. That is exactly why it is read next to the other iron markers rather than alone.

Vitamin D

Vitamin D deficiency is common, easy to measure, and easy to correct, which makes it worth checking early.

A systematic review found that people with fibromyalgia tend to have lower vitamin D levels than healthy controls, while also finding conflicting results on whether supplementation improves pain or symptom control, with no clear consensus on its role in management (PMID 30886978). A small randomized placebo-controlled trial in 30 women with fibromyalgia and baseline 25-hydroxyvitamin D below 32 ng/mL supplemented to a target range of 32 to 48 ng/mL and found a significant group effect on pain scores over the treatment period, while explicitly calling for larger studies before drawing firm conclusions (PMID 24438771).

So: association is reasonably well supported, treatment benefit is promising but not established. That is the accurate summary, and anyone who tells you vitamin D fixes fibromyalgia is ahead of the evidence.

What to ask for: 25-hydroxyvitamin D, sometimes written 25(OH)D. This is the storage form and the one that reflects your status. Ask specifically for 25-hydroxy rather than 1,25-dihydroxy, which answers a different clinical question.

The number you will argue about. A 2011 Endocrine Society guideline defined vitamin D deficiency as 25(OH)D below 20 ng/mL and insufficiency as 21 to 29 ng/mL, recommending a target above 30 ng/mL for patients at risk (PMID 21646368). The Endocrine Society's 2024 guideline took a notably more conservative position, declining to recommend routine 25(OH)D testing in healthy adults and stepping back from specific target thresholds in the general population (PMID 38828931). Both are real guidelines from the same body, thirteen years apart, and they disagree. If your doctor's threshold differs from something you read online, that gap is where it comes from, and it is a reasonable thing to raise directly.

Vitamin B12 and Folate

B12 deficiency produces fatigue, cognitive fog, numbness and tingling, and mood changes. It is one of the more satisfying findings in this whole workup because it is correctable.

The catch is that serum B12 alone is an imperfect test. There is a recognized borderline zone in which a serum B12 result cannot settle the question on its own, and in that zone, adding methylmalonic acid (MMA) or homocysteine gives you a functional readout of whether your cells are actually able to use the B12 that is circulating (PMID 24942828). This is the same logic that shows up repeatedly in this guide: a number inside the reference range is not automatically a number doing its job.

What to ask for: serum B12, serum folate, and, if your B12 lands in a borderline range or your symptoms are strongly suggestive, methylmalonic acid and homocysteine.

A word about homocysteine specifically, because it gets waved off. It is a methylation-cycle marker. It rises when B12, folate, or B6 run short, and also with thyroid or kidney trouble. In my read it is one of the more consequential numbers on a panel, because an elevated result is telling you the methylation machinery is running short somewhere findable. The way to act on it is not to chase the number with blind supplementation. Find which input is short, correct that, and let homocysteine confirm the fix on a retest. My functional target of 5 to 7 in the table below reflects how seriously I take it. It is a window, not a floor. Lower is not better: a homocysteine driven below that window points toward overmethylation and B vitamin intolerance, which is its own problem rather than a better result.

A medication note, and please read the second half of it. Metformin use in type 2 diabetes is associated with lower vitamin B12 levels in a systematic review and meta-analysis (PMID 27130885). If you take metformin, or any other medication, and you are concerned about a lab value, the action is to bring it to your prescriber. It is never to stop or change the medication yourself. Monitoring and supplementation are decisions that belong to the person who prescribed the drug.

Section 2: The Rule-Out Tier, Part Two (Inflammatory Screens, Celiac, Sleep)

Inflammatory and Autoimmune Screens

This is the part of the workup where more testing can actively make things worse, so it is worth understanding before you ask for anything.

Fibromyalgia is not an inflammatory arthritis. It does not damage joints and it does not typically produce the inflammatory markers that rheumatoid arthritis or lupus produce. But those conditions can look similar from the outside early on, and they can also occur alongside fibromyalgia, so a screen is reasonable when the clinical picture warrants it.

Erythrocyte sedimentation rate (ESR) and C-reactive protein (CRP) are the general inflammation markers. In fibromyalgia they are usually normal or near normal. A clearly and persistently elevated ESR or CRP is a signal that something else is going on and deserves investigation. This matters especially if you are over 50 with new shoulder and hip girdle pain and stiffness, because polymyalgia rheumatica is a distinct condition whose classification criteria are built around exactly that presentation plus elevated inflammatory markers, and it responds to entirely different treatment (PMID 22388996).

Rheumatoid factor (RF), anti-CCP antibodies, and antinuclear antibody (ANA) are the more specific autoimmune screens, and they come with a real warning. A review of consecutive rheumatology referrals found that at least half of the ANA, RF, ENA, and anti-dsDNA testing performed before referral was ordered inappropriately, with 20 percent of ANA testing done with no clinical indication at all, and noted that RF and ANA have low positive predictive value in general practice (PMID 32522598). Translated: if you order these tests without a clinical reason, you will collect positive results that mean nothing, and those results generate anxiety, repeat testing, and referrals that lead nowhere.

So how should you use this? Ask your physician whether your specific picture warrants autoimmune screening, and accept a well-reasoned no. If your exam shows no synovitis, no rash, no dry eyes and mouth, no Raynaud's, and normal inflammatory markers, a shotgun autoimmune panel is more likely to confuse your chart than clarify it. If you do have those features, the screening is appropriate and you should push for it.

And if a screen comes back positive: that finding belongs in front of a physician, ideally a rheumatologist, before you draw any conclusions. A low-titer positive ANA is common in the healthy population. It is not a diagnosis.

Celiac Disease and Gluten

Celiac disease can present with musculoskeletal and other symptoms outside the gut, which is part of why it is under-recognized (PMID 31513026). Given how much overlap exists between celiac symptoms and fibromyalgia symptoms, screening is reasonable, and it is a simple blood test.

What to ask for: tissue transglutaminase IgA (tTG-IgA) together with total serum IgA. The total IgA matters because selective IgA deficiency will make the tTG-IgA falsely negative, and current guidance is to test both together (PMID 36602836).

Critical practical point that trips people up constantly: celiac serology is only valid if you are actively eating gluten. If you have already gone gluten free, the test can come back negative regardless of whether you have celiac disease. Guideline guidance is that patients should be on a gluten-containing diet when tested (PMID 36602836). If you have already stopped gluten and want to be tested properly, that is a conversation to have with your physician before you change anything, because a gluten challenge is a medical decision with real symptom consequences.

What about gluten sensitivity without celiac disease? This is a genuinely open area. A case series described 20 fibromyalgia patients without celiac disease who improved on a gluten-free diet, with all of them showing intraepithelial lymphocytosis without villous atrophy on biopsy, and the authors framed it as support for a hypothesis rather than as proof (PMID 24728027). That is a case series, not a controlled trial, and it should be read as a reason for curiosity and a supervised trial, not as an established mechanism.

Testing here deserves a precise answer, because it usually gets a sloppy one. Standard celiac serology is narrow by design. It is built to answer one question, which is whether you have celiac disease, and it is not built to see reactivity outside that question. Specialty laboratories offer considerably wider panels that measure antibodies against individual wheat and gluten peptides rather than against the whole protein, alongside related targets such as non-gluten wheat proteins and the transglutaminase enzymes. The Wheat Zoomer from Vibrant Wellness and Array 3X from Cyrex Laboratories are the two most commonly used in functional medicine. Those are real measurements of real antibodies, run at a level of detail standard celiac testing does not attempt, and they can surface reactivity that a normal tTG-IgA will never show you.

What is not established is the step from a positive peptide result to a diagnosis. The current expert consensus definition of non-celiac gluten sensitivity, known as the Salerno criteria, is built entirely on symptom response: celiac disease and wheat allergy ruled out first, then symptoms that improve when gluten is withdrawn and return when it is reintroduced, ideally under blinded conditions (PMID 26096570). There is no antibody cutoff anywhere in that definition, because no antibody has been validated as the thing that settles it.

So the honest way to use one of these panels is as information that helps you and your clinician decide whether a careful elimination and reintroduction is worth doing, and as a baseline you can retest later to watch a direction of travel. It is a reason to run the trial, not a substitute for it. A practitioner who tells you a peptide panel diagnoses gluten sensitivity is going further than the evidence goes. A practitioner who tells you nothing can be measured at all is behind where the laboratories actually are. Both statements are wrong, and the space between them is where the useful work happens.

Sleep, Which Is Not a Blood Test but Belongs Here Anyway

Unrefreshing sleep is one of the core features of fibromyalgia, and it is also the core feature of obstructive sleep apnea. A systematic review and meta-analysis found that approximately 21 percent of patients with obstructive sleep apnea-hypopnea syndrome met criteria for fibromyalgia (PMID 38831795).

You cannot sort this out with a blood draw. If you snore, wake unrefreshed, wake gasping, have witnessed apneas, have morning headaches, or fall asleep during the day, ask your physician about a sleep study. Treating sleep apnea will not cure fibromyalgia. But leaving untreated apnea in place while you work on everything else means you are trying to repair a system that never gets to run its overnight repair cycle.

Section 3: The Terrain Tier (Blood Sugar, Inflammation, Stress Axis)

Everything above is about ruling things out. This section is about describing the environment your nervous system is working in. These are not fibromyalgia tests. They are levers.

Blood Sugar and Insulin

Insulin resistance is worth looking at here, with the caveat that this specific literature requires care.

An observational study reported an association between insulin resistance, measured by HbA1c, and central pain in patients with fibromyalgia (PMID 33740353). You should also know that an earlier paper from the same group, titled "Is insulin resistance the cause of fibromyalgia? A preliminary report," has been **retracted** and is not cited anywhere in this guide. I am telling you that explicitly because you are likely to find that retracted paper circulating in fibromyalgia communities online, presented as evidence that insulin resistance causes fibromyalgia. It is not usable evidence. The association is worth investigating. The causal claim is not supported.

What to ask for: fasting glucose, HbA1c, and fasting insulin. Fasting insulin is the one that usually gets left off and is often the most informative of the three, because insulin can climb for years while fasting glucose still reads normal. HOMA-IR is not a separate test; it is a calculation your clinician can run from your fasting glucose and fasting insulin.

What a result actually tells you. Rising fasting insulin with normal glucose describes a metabolic pattern worth addressing through nutrition, movement, and sleep. It does not diagnose fibromyalgia, it does not explain your pain by itself, and it is not a reason to panic. It is a modifiable finding, which is exactly what makes it worth measuring.

Inflammatory Markers, Read Carefully

Fibromyalgia is not classically an inflammatory disease, but there is evidence of low-grade inflammatory signal in at least a subset of patients.

A study of 105 fibromyalgia patients and 61 healthy controls found high-sensitivity CRP levels only marginally higher in the fibromyalgia group overall, but with a significantly higher proportion of abnormal values, and the hsCRP values correlated with body mass index, ESR, IL-6, and IL-8. Notably, ESR, IL-6, and IL-8 did not themselves differ between the groups (PMID 23124693). A more recent systematic review and meta-analysis of 18 observational studies found elevated hsCRP in fibromyalgia relative to controls as the one biomarker reaching statistical significance in quantitative synthesis, while hair cortisol, IL-6, and interferon gamma showed no significant between-group differences, and the authors explicitly noted moderate methodological quality, high heterogeneity, and limited direct clinical applicability (PMID 42477737). At the time of writing this reference is published online ahead of print.

How to use that. hs-CRP is a nonspecific screening marker. An elevated result says an inflammatory load exists somewhere. It does not say where that load is coming from, and it confirms nothing on its own. That makes it worth knowing and worth addressing, because inflammatory load is often modifiable, but it is not a fibromyalgia marker, it does not track your pain reliably, and it correlates strongly with body mass index, which means it is telling you about your metabolic terrain as much as anything else.

The Stress Axis (HPA Axis and Cortisol)

The hypothalamic-pituitary-adrenal axis is the system that governs your cortisol rhythm, your stress response, and a good deal of your energy and sleep regulation. Disturbances in this axis have been described in fibromyalgia since at least the mid 1990s (PMID 7980669).

Here is the part that most functional medicine marketing leaves out, and that you deserve to have: the relationship between cortisol findings and fibromyalgia pain is not clean. A study of salivary cortisol release and HPA feedback sensitivity in fibromyalgia found that the alterations tracked with depression rather than with pain (PMID 20627822). And the meta-analysis referenced above found no significant between-group difference in hair cortisol (PMID 42477737).

What that means for you practically. Cortisol testing, whether salivary, urinary, or serum, can be genuinely informative about your stress physiology, your sleep-wake rhythm, and your energy pattern across a day. It is a reasonable thing to run when fatigue, sleep disruption, and stress load dominate your picture. What it is not is a fibromyalgia test, and a "flat cortisol curve" result is not a diagnosis of anything. Treat it as one description of your terrain, interpreted next to everything else, by a clinician.

Also note: true adrenal insufficiency is a serious medical condition diagnosed with specific testing by a physician, and it is a completely different thing from the cortisol rhythm patterns discussed here. If you have unexplained weight loss, low blood pressure, salt craving, darkening of the skin, or episodes of collapse, that needs a physician now.

Magnesium and Other Nutrients

Magnesium is one of the most frequently discussed nutrients in fibromyalgia, and the honest state of the evidence is thinner than the internet suggests. An observational cross-sectional study reported an association between serum magnesium levels and both sleep quality and disease severity in fibromyalgia (PMID 40696633). That is a single cross-sectional study, and association is not causation.

There is also a real technical problem worth knowing: serum magnesium reflects only a small fraction of total body magnesium and can look normal when tissue stores are not. That is a recognized limitation of the test rather than a reason to distrust the lab.

Practical takeaway. Serum magnesium is inexpensive and can be added to a panel. Read a low result as meaningful and a normal result as inconclusive rather than reassuring. Any decision about supplementation belongs with your clinician, particularly if you have kidney disease, where magnesium supplementation carries real risk.

Small Fiber Neuropathy

A meaningful subset of people diagnosed with fibromyalgia have evidence of small fiber neuropathy, which can be assessed by measuring intraepidermal nerve fiber density on a skin punch biopsy (PMID 33802768). This matters if your symptoms include prominent burning, electric, or shooting pain, or numbness and tingling, on top of the widespread aching.

This is not a routine test and it is not something to request casually. It is a conversation to have with a neurologist or a physician who works with nerve symptoms, and the reason to have it is that a small fiber neuropathy finding can change the workup, pointing toward causes such as diabetes, B12 deficiency, or autoimmune conditions that have their own specific management.

Section 4: "In Range" Versus "Optimal," Explained Honestly

This is the section where you will hear two credible people say different things, so let us be precise about what each one is actually claiming.

What a laboratory reference range is. It is a statistical description of results from a reference population, usually constructed so that roughly the middle 95 percent of that population falls inside it. It is a description of what is common. It was never designed to describe what is ideal, and it is not the same thing as a treatment threshold.

What "optimal range" means in functional medicine. Many functional medicine practitioners, myself included, work with narrower target windows than the laboratory reference range for certain markers, on the reasoning that being at the bottom edge of a broad population range is not the same as being where a given system functions best, and that catching a drift early is better than waiting for it to cross a diagnostic line.

Here is the honest part. For most markers, those narrower windows are a clinical framework, not a validated diagnostic threshold. They come from clinical experience, from physiological reasoning, and from patterns practitioners observe over many patients. They are not, for the most part, numbers that a randomized trial has established as the point where symptoms improve. Anyone presenting a functional optimal range as settled science is overstating it, and you should be skeptical of a guide that does not tell you that.

Where numbers in this guide do come from published sources, I have said so explicitly:

- The vitamin D thresholds of below 20 ng/mL for deficiency and 21 to 29 ng/mL for insufficiency come from a 2011 Endocrine Society guideline (PMID 21646368), and the same body's 2024 guideline moved away from routine testing and specific general-population targets (PMID 38828931).
- The 32 to 48 ng/mL vitamin D target used in the fibromyalgia randomized trial described earlier is that trial's protocol target, in 30 women, not a general recommendation (PMID 24438771).
- The ferritin figure of below 50 ng/mL comes from a single case-control study's finding of increased odds of fibromyalgia at that level (PMID 20087382), and it sits alongside a study that found no iron differences at all (PMID 22095117) and a systematic review indicating that ferritin cutoffs for deficiency are themselves contested (PMID 34028001).

The functional targets I use. Rather than leave you guessing, here are the actual target windows from my blood chemistry framework for the markers this guide covers. Read the caution above first, because it applies to every number in this table: these are clinical targets drawn from functional medicine practice, not validated diagnostic thresholds, and being outside one of them is a question to raise, never a diagnosis to make.

Marker	Standard lab range is roughly	Root Health functional target
TSH	0.4 to 4.0 mIU/L	1.0 to 2.5 mIU/L
Free T4	0.7 to 1.7 ng/dL	0.9 to 1.4 ng/dL
Free T3	2.0 to 4.2 pg/mL	2.8 to 3.6 pg/mL
Reverse T3	up to 24 ng/dL	up to 15 ng/dL
TPO antibodies	up to 35 IU/mL	up to 9 IU/mL
Thyroglobulin antibodies	up to 20 IU/mL	up to 1 IU/mL
Ferritin	25 to 300 ng/mL	50 to 150 ng/mL
Transferrin saturation	15 to 45 percent	20 to 40 percent
Vitamin D (25-OH)	20 to 100 ng/mL	40 to 80 ng/mL
Vitamin B12	200 to 1100 pg/mL	500 to 900 pg/mL
Folate (RBC)	400 to 2000 ng/mL	600 to 1500 ng/mL
Methylmalonic acid	up to 243 nmol/L	up to 162 nmol/L
Homocysteine	up to 10 umol/L	5 to 7 umol/L
hs-CRP	up to 1.0 mg/L	up to 0.5 mg/L
ESR	up to 25 mm/hr	up to 15 mm/hr
Fasting glucose	70 to 99 mg/dL	75 to 86 mg/dL
HbA1c	4.0 to 5.6 percent	4.6 to 5.3 percent
Fasting insulin	2.0 to 10.0 uIU/mL	2.6 to 5.0 uIU/mL
HOMA-IR	up to 2.5	0.5 to 1.4
Magnesium (serum)	1.5 to 2.7 mg/dL	2.0 to 2.5 mg/dL
Magnesium (RBC)	4.0 to 7.2 mg/dL	5.2 to 6.8 mg/dL
Cortisol (morning)	6.0 to 25.0 ug/dL	12.0 to 22.0 ug/dL

Marker	Standard lab range is roughly	Root Health functional target
DHEA-S	75 to 450 ug/dL	125 to 350 ug/dL

A few notes on using this table well. Reference ranges printed on your own report may differ from the middle column, because each laboratory sets its own from its own population, so always compare against the range on your actual result. Hemoglobin, hematocrit, and several hormone targets vary by sex and by age, so those are left to your clinician rather than reduced to one line here. And a result sitting just outside a functional target while comfortably inside the standard range is common, is not an emergency, and is exactly the kind of finding that deserves a conversation rather than a supplement purchase.

How to use the functional lens well. Look at where your result sits inside the range, not just whether it is inside. Look at the direction of travel across multiple draws over time, which is far more informative than any single value. Read markers in groups rather than one at a time. And treat a “low normal” result as a question worth asking rather than as an answer. That is the useful core of the functional approach, and none of it requires believing an unvalidated cutoff.

Section 5: Reading Your Results as a Pattern

Individual values are noisy. Patterns are where the information lives. Here are combinations worth noticing, all of which are conversation starters with your clinician rather than conclusions.

Everything normal. This is the most common result and it is genuinely good news. It means the correctable mimics have been checked. From here, the work moves to the terrain tier and to the parts of care that do not depend on labs at all, including sleep, graded movement, nervous system regulation, and nutrition.

Low or low-normal ferritin plus poor sleep plus restless legs. Iron status and sleep quality are worth looking at together rather than separately. This is a correctable-in-principle combination and worth raising specifically.

Positive thyroid antibodies with a normal TSH. Autoimmune thyroid activity without current gland failure. Not a fibromyalgia explanation, but a real finding that warrants monitoring and physician follow-up.

Elevated hs-CRP with a raised body mass index. The published correlation between these two is strong (PMID 23124693), which points the conversation toward metabolic and inflammatory terrain rather than toward an autoimmune diagnosis.

Elevated ESR or CRP that stays elevated. This is the pattern that should push hardest toward further physician workup, because persistent inflammatory marker elevation is not typical of fibromyalgia alone and points toward something additional going on.

Rising fasting insulin with a normal fasting glucose. A metabolic pattern that is modifiable, worth addressing, and not by itself an explanation for widespread pain.

Borderline serum B12 with symptoms that fit. The situation where methylmalonic acid or homocysteine earns its place (PMID 24942828).

A positive autoimmune screen of any kind. Stop interpreting and get a rheumatology opinion. This is the one pattern in the list where self-interpretation carries real cost.

Section 6: The Conversation With Your Doctor

Most of the tests in this guide are ordinary, inexpensive, and orderable by any primary care provider. You do not need a specialist and you do not need a functional medicine practice to get the rule-out tier drawn. What you often do need is to ask specifically, because the default panel is narrower than what is described here.

How to open the conversation

Something like this works well, and it works because it is true:

> *"I understand fibromyalgia is diagnosed clinically and that there is no lab test that confirms it. What I want to do is make sure we have ruled out the treatable conditions that can look like this or make it worse. Can we check thyroid function including antibodies, iron status including ferritin, vitamin D, B12, celiac screening, and basic inflammatory markers?"*

That framing does three things. It shows you understand the diagnosis rather than shopping for a different one. It names rule-out, which is a goal your physician already shares. And it asks for specific tests instead of asking for "everything."

A request list you can bring

Tier one, the rule-out round. Complete blood count with differential. Comprehensive metabolic panel. TSH, free T4, free T3, TPO antibodies, thyroglobulin antibodies. Ferritin, serum iron, TIBC, transferrin saturation. 25-hydroxyvitamin D. Vitamin B12 and folate. Tissue transglutaminase IgA with total IgA. ESR and hs-CRP. Creatine kinase if you have prominent muscle weakness as opposed to pain.

Tier two, added when the picture warrants it. Methylmalonic acid and homocysteine if B12 is borderline. Fasting insulin alongside fasting glucose and HbA1c. Magnesium. Autoimmune screening (ANA, RF, anti-CCP) only if there are clinical features that justify it, for the reasons in Section 2. A sleep study if any sleep apnea features are present. Cortisol rhythm assessment if fatigue and stress physiology dominate. Referral for small fiber neuropathy evaluation if burning, electric, or numb-and-tingling symptoms are prominent.

If your request is declined

Ask why, and listen to the answer, because sometimes it is a good one. "Your inflammatory markers are normal and you have no synovitis, so an ANA is more likely to mislead us than help" is sound medicine, not dismissal. "There is no point testing anything because it is just fibromyalgia" is a different kind of answer, and it is a reasonable trigger to seek a second opinion.

You can also ask your physician to document the request and the reasoning in your chart. That is a normal request, it costs nothing, and it tends to make the conversation more precise.

Practical notes on getting drawn

Fast for 8 to 12 hours if fasting glucose, fasting insulin, or a lipid panel is included, and drink water. If you take biotin, including in a hair, skin, and nail supplement, tell the lab and your physician, because high-dose biotin can interfere with several immunoassays including thyroid tests. Ask for a copy of your own results, in numbers, not just a “normal” phone call. You are entitled to them and you cannot track a trend you never saw.

Section 7: What These Labs Cannot Tell You

Being clear about the limits is part of respecting your time and your money.

They cannot confirm fibromyalgia. Nothing in this guide does that, and any test marketed as a fibromyalgia diagnostic should be approached with real skepticism.

They cannot measure your pain. There is no lab value that tracks how much you hurt. Your reported experience is the primary data on that, and it is legitimate data.

They cannot rank your root causes for you. Several findings often show up at once. Deciding which one to address first, and in what order, is clinical judgment applied to your specific history. That is the work a good clinician does and it is not something a document can do for you.

They cannot substitute for the interventions that actually have the strongest evidence in fibromyalgia, which are not laboratory findings at all. Graded physical activity, sleep, and nervous system regulation carry more weight in the outcome literature than any marker in this guide, and no amount of testing replaces them.

Normal labs do not mean nothing is wrong. Say that one out loud a few times if you need to. It is the single most common misreading in this entire condition, and it has kept a lot of people from getting care they needed.

Section 8: When to Seek Clinical Review

Do not wait on lab results, and do not work through this guide first, if you have any of the following:

- Fever, drenching night sweats, or unexplained weight loss alongside your pain
- Visible joint swelling, warmth, or redness, particularly with morning stiffness lasting more than an hour
- New or worsening muscle weakness, as distinct from pain or fatigue
- New onset of widespread pain and girdle stiffness after age 50, particularly with new headache, jaw pain when chewing, scalp tenderness, or any visual change, which requires urgent evaluation
- Progressive numbness, foot drop, loss of balance, or falls
- Any bladder or bowel control change, or numbness in the saddle region, which is an emergency
- Thoughts of harming yourself, which warrants immediate help through your physician, a crisis line, or emergency services

Fibromyalgia is a real diagnosis, and it is also a diagnosis that can sit on top of something else. Getting a new or changing symptom evaluated is not being difficult. It is good self-advocacy.

Your Next Step

Take this guide, or just the request list in Section 6, to your next appointment. Get the rule-out round drawn. Ask for your results in numbers.

Then bring those numbers to someone who will read them as a pattern rather than as a pass or fail. If you want that conversation with me, a Root Discovery Session is complimentary and exists for exactly this purpose:

> ***Book a free Root Discovery Session: rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=fibromyalgia-labs***

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

If you would rather start on your own, your own record keeping is where the information lives. Take the request list in Section 6 to your own clinician, and start recording your results and symptoms over time. A single draw is a snapshot. The direction of travel across several draws is where the information actually lives, and no one will keep that record for you:

> ***Free symptom tracker: rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=fibromyalgia-labs***

And if you want a second set of eyes on the pattern you are seeing, write to me. A second opinion on how to read a trend, and help deciding what is worth asking for next, are often the fastest way to stop spinning on results nobody has explained to you:

> ***Get in touch: rootcauseunlocked.com/contact?utm_source=guide&utm_medium=ebook&utm_campaign=fibromyalgia-labs***

References

Every paper below was pulled and read against its original record, and checked to be sure it had not been retracted. One paper that came up in the research had been, and instead of quietly leaving it out I named it in the blood sugar section, because it still circulates in fibromyalgia groups as proof of something it never proved.

1. Clauw DJ. Fibromyalgia: a clinical review. JAMA. 2014 Apr 16;311(15):1547-55. PMID: 24737367

2. Wolfe F, Clauw DJ, Fitzcharles MA, Goldenberg DL, Häuser W, Katz RL, et al. 2016 Revisions to the 2010/2011 fibromyalgia diagnostic criteria. Semin Arthritis Rheum. 2016 Dec;46(3):319-329. PMID: 27916278

3. Haliloglu S, Carlioglu A, Akdeniz D, Karaaslan Y, Kosar A. Fibromyalgia in patients with other rheumatic diseases: prevalence and relationship with disease activity. Rheumatol Int. 2014 Sep;34(9):1275-80. PMID: 24589726

4. Bazzichi L, Rossi A, Zirafa C, Monzani F, Tognini S, Dardano A, et al. Thyroid autoimmunity may represent a predisposition for the development of fibromyalgia? Rheumatol Int. 2012 Feb;32(2):335-41. PMID: 21085966

5. Ortancil O, Sanli A, Eryuksel R, Basaran A, Ankarali H. Association between serum ferritin level and fibromyalgia syndrome. Eur J Clin Nutr. 2010 Mar;64(3):308-12. PMID: 20087382

6. Mader R, Koton Y, Buskila D, Herer P, Elias M. Serum iron and iron stores in non-anemic patients with fibromyalgia. Clin Rheumatol. 2012 Apr;31(4):595-9. PMID: 22095117

7. Boomershine CS, Koch TA, Morris D. A Blinded, Randomized, Placebo-Controlled Study to Investigate the Efficacy and Safety of Ferric Carboxymaltose in Iron-Deficient Patients with Fibromyalgia. Rheumatol Ther. 2018 Jun;5(1):271-281. PMID: 29149437

8. Garcia-Casal MN, Pasricha SR, Martinez RX, Lopez-Perez L, Peña-Rosas JP. Serum or plasma ferritin concentration as an index of iron deficiency and overload. Cochrane Database Syst Rev. 2021 May 24;5(5):CD011817. PMID: 34028001

9. Ellis SD, Kelly ST, Shurlock JH, Hepburn ALN. The role of vitamin D testing and replacement in fibromyalgia: a systematic literature review. BMC Rheumatol. 2018;2:28. PMID: 30886978

10. Wepner F, Scheuer R, Schuetz-Wieser B, Machacek P, Pieler-Bruha E, Cross HS, et al. Effects of vitamin D on patients with fibromyalgia syndrome: a randomized placebo-controlled trial. Pain. 2014 Feb;155(2):261-268. PMID: 24438771

11. Holick MF, Binkley NC, Bischoff-Ferrari HA, Gordon CM, Hanley DA, Heaney RP, et al. Evaluation, treatment, and prevention of vitamin D deficiency: an Endocrine Society clinical practice guideline. J Clin Endocrinol Metab. 2011 Jul;96(7):1911-30. PMID: 21646368

12. Demay MB, Pittas AG, Bikle DD, Diab DL, Kiely ME, Lazaretti-Castro M, et al. Vitamin D for the Prevention of Disease: An Endocrine Society Clinical Practice Guideline. J Clin Endocrinol Metab. 2024 Jul 12;109(8):1907-1947. PMID: 38828931

13. Devalia V, Hamilton MS, Molloy AM. Guidelines for the diagnosis and treatment of cobalamin and folate disorders. Br J Haematol. 2014 Aug;166(4):496-513. PMID: 24942828

14. Chapman LE, Darling AL, Brown JE. Association between metformin and vitamin B12 deficiency in patients with type 2 diabetes: A systematic review and meta-analysis. Diabetes Metab. 2016 Nov;42(5):316-327. PMID: 27130885

15. Dasgupta B, Cimmino MA, Maradit-Kremers H, Schmidt WA, Schirmer M, Salvarani C, et al. 2012 provisional classification criteria for polymyalgia rheumatica: a European League Against Rheumatism/American College of Rheumatology collaborative initiative. Ann Rheum Dis. 2012 Apr;71(4):484-92. PMID: 22388996

16. Ahrari A, Barrett SS, Basharat P, Rohekar S, Pope JE. Appropriateness of laboratory tests in the diagnosis of inflammatory rheumatic diseases among patients newly referred to rheumatologists. Joint Bone Spine. 2020 Dec;87(6):588-595. PMID: 32522598

17. Therrien A, Kelly CP, Silvester JA. Celiac Disease: Extraintestinal Manifestations and Associated Conditions. J Clin Gastroenterol. 2020 Jan;54(1):8-21. PMID: 31513026

18. Rubio-Tapia A, Hill ID, Semrad C, Kelly CP, Greer KB, Limketkai BN, et al. American College of Gastroenterology Guidelines Update: Diagnosis and Management of Celiac Disease. Am J Gastroenterol. 2023 Jan 1;118(1):59-76. PMID: 36602836

19. Isasi C, Colmenero I, Casco F, Tejerina E, Fernandez N, Serrano-Vela JI, et al. Fibromyalgia and non-celiac gluten sensitivity: a description with remission of fibromyalgia. Rheumatol Int. 2014 Nov;34(11):1607-12. PMID: 24728027

20. Catassi C, Elli L, Bonaz B, Bouma G, Carroccio A, Castillejo G, et al. Diagnosis of Non-Celiac Gluten Sensitivity (NCGS): The Salerno Experts' Criteria. Nutrients. 2015 Jun 18;7(6):4966-77. PMID: 26096570

21. He J, Chen M, Huang N, Wang B. Fibromyalgia in obstructive sleep apnea-hypopnea syndrome: a systematic review and meta-analysis. Front Physiol. 2024;15:1394865. PMID: 38831795

22. Pappolla MA, Manchikanti L, Candido KD, Grieg N, Seffinger M, Ahmed F, et al. Insulin Resistance is Associated with Central Pain in Patients with Fibromyalgia. Pain Physician. 2021 Mar;24(2):175-184. PMID: 33740353

23. Xiao Y, Haynes WL, Michalek JE, Russell IJ. Elevated serum high-sensitivity C-reactive protein levels in fibromyalgia syndrome patients correlate with body mass index, interleukin-6, interleukin-8, erythrocyte sedimentation rate. Rheumatol Int. 2013 May;33(5):1259-64. PMID: 23124693

24. Fernández-Rodríguez FJ, Delgado-Pérez E, Salgado-Ortiz CS, Quishpe-Rivera EA, Sosa-Reina MD. Inflammatory biomarker alterations in patients with fibromyalgia: a systematic review and meta-analysis. BMC Rheumatol. 2026 Jul 20. Online ahead of print. PMID: 42477737

25. Crofford LJ, Pillemer SR, Kalogeras KT, Cash JM, Michelson D, Kling MA, et al. Hypothalamic-pituitary-adrenal axis perturbations in patients with fibromyalgia. Arthritis Rheum. 1994 Nov;37(11):1583-92. PMID: 7980669

26. Wingenfeld K, Nutzinger D, Kauth J, Hellhammer DH, Lautenbacher S. Salivary cortisol release and hypothalamic pituitary adrenal axis feedback sensitivity in fibromyalgia is associated with depression but not with pain. J Pain. 2010 Nov;11(11):1195-202. PMID: 20627822

27. Alisik T, Reis Altan YC, Olkay SG, Sahingoz Bakirci E. Serum magnesium levels and their association with sleep quality and disease severity in fibromyalgia syndrome: An observational cross-sectional study. Medicine (Baltimore). 2025 Jul 18;104(29):e43446. PMID: 40696633

28. Kelley MA, Hackshaw KV. Intraepidermal Nerve Fiber Density as Measured by Skin Punch Biopsy as a Marker for Small Fiber Neuropathy: Application in Patients with Fibromyalgia. Diagnostics (Basel). 2021 Mar 17;11(3):536. PMID: 33802768

Working with Root Health

This guide is general education by design, written so it is useful without knowing anything about your labs. Personalized work is different: it starts with testing, your history, your medication list, and the symptoms costing you the most, then sequences changes instead of piling them on.

If you want that, a Root Discovery Session is free and is the right first step. Book at rootcauseunlocked.com/discovery.

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

If you would rather start on your own, the symptom tracker at rootcauseunlocked.com/tracker is free and works anywhere. In fibromyalgia it does real work, because the changes are gradual and memory is an unreliable instrument across weeks.

Reach me at (919) 662-0520 or at 503 US-70 East, Suite L, Garner, NC 27529.

Root Health | Dr. Adam Seay, DC | Garner, NC | rootcauseunlocked.com

> ***Please read this first***
>
> *This guide is educational. It is not medical advice, it does not diagnose anything, and it does not replace your own clinician. No eating pattern described here treats, cures, or reverses fibromyalgia. Food can support how you feel. It is not a substitute for care.*
>
> ***Talk with your prescriber or a registered dietitian before making major dietary changes****, particularly if you have diabetes or take medication that lowers blood sugar, take a blood thinner (vitamin K intake matters), have kidney or liver disease, are pregnant or nursing, have celiac disease, or have any history of an eating disorder. Restrictive eating is not a neutral intervention.*
>
> ***If you have not been tested for celiac disease, do not remove gluten before testing.*** *Going gluten-free first can make celiac testing come back falsely normal, and celiac disease is a real diagnosis with real consequences that you want identified properly.*
>
> *Unexplained weight loss, difficulty swallowing, blood in the stool, persistent vomiting, or nighttime symptoms that wake you are not diet-experiment territory. They need a clinician.*

What you are actually working toward

Not a “clean eating” identity. Not a moral scorecard about food. Something much more specific and much more useful:

Steadier energy across the afternoon instead of a wall at two o’clock. Fewer of the nights where your gut keeps you awake and then your pain is worse the next day. A body that is less reactive to ordinary meals. Fewer flare days in a month than you had before.

Food will not do this alone. It also will not do nothing.

What the evidence actually shows

I want to be honest with you here, because most fibromyalgia diet content is not.

There is no proven fibromyalgia diet. A systematic review and best-evidence synthesis of 22 studies covering 17 different nutritional interventions found signals for several approaches, including vegan eating and low FODMAP eating, and concluded that the study quality was frequently poor, the heterogeneity wide, the samples small, and the bias high, leaving insufficient evidence to recommend any one nutritional intervention for fibromyalgia (PMID 32878326). A second systematic review of seven clinical trials reached a similar place: pain and function improved in five of seven studies, and the statistical quality was poor enough that firm conclusions were not possible (PMID 30735059).

So what is the point of a diet guide?

Two things the reviews also show. First, the direction of effect across these small studies is mostly favorable, and the interventions themselves are low risk compared with most alternatives. Second, several of the mechanisms that food touches are genuinely part of the fibromyalgia picture: gut symptoms, sleep, body weight, blood sugar stability, and the composition of the gut microbiome (PMID 38279728).

That is what makes food worth your attention. Not because it is a cure, but because it is a lever you control every day, at low risk, with a reasonable chance of a modest gain.

Here is how I want you to hold it: **eating well is the foundation, and the specific experiments sit on top of it.** Do not start with the experiments.

Part one: the foundation

If you do nothing else in this guide, do this part. It is the most supported and the least restrictive, and it is the base every other section assumes.

The pattern, described plainly

A Mediterranean-style, mostly whole-food pattern. In practice:

- **Vegetables and fruit at most meals.** Aim for color variety across a week rather than perfection in a day.
- **Olive oil as your main fat.** In a preliminary randomized study, women with fibromyalgia who consumed extra virgin olive oil daily for three weeks showed improvements in markers of oxidative stress along with Fibromyalgia Impact Questionnaire and mental health summary scores compared with a refined olive oil group (PMID 27443526). Small study, short duration, but an easy and pleasant change to make.
- **Fish two or three times a week**, especially oily fish (salmon, sardines, mackerel, trout).
- **Legumes, nuts, and seeds** as regular protein and fiber sources, adjusted if you are also doing the FODMAP work below.
- **Whole grains** in reasonable portions if you tolerate them.
- **Poultry and eggs** in moderation, red meat less often.
- **Much less ultra-processed food.** More on that in a moment.

Why this pattern and not another

In a study of 100 outpatients with fibromyalgia, a personalized Mediterranean diet followed for eight weeks was associated with improvement in most fibromyalgia parameters, including disability scores, fatigue, and anxiety, compared with a general balanced diet (PMID 38683449). A systematic review of Mediterranean diet studies in fibromyalgia found that adherence was associated with improvements in widespread pain, fatigue, sleep disturbance, and cognitive symptoms, while describing the likely effect size as low to moderate (PMID 40576703). Broader reviews of chronic pain nutrition arrive at a similar structure: an eating pattern built around anti-inflammatory and antioxidant-rich whole foods rather than any single food or nutrient (PMID 29679994).

The ultra-processed food question

A case-control study comparing 45 adults with fibromyalgia to 44 healthy controls found that the fibromyalgia group got a significantly higher share of total energy from ultra-processed foods (34.5% versus 26.7%), scored higher on a dietary inflammatory index, and had lower intakes of magnesium, vitamin C, and polyphenols alongside higher saturated fat and refined carbohydrate intake (PMID 41021618).

Read the study design honestly: this shows an association, not that ultra-processed food caused anyone's fibromyalgia. But the practical takeaway holds up regardless of direction. Packaged food crowds out the nutrients the rest of this guide depends on. If you want one change with the best ratio of benefit to effort, it is cooking more of your own food, even simply.

What “cooking more” can honestly look like

Fibromyalgia fatigue is real, and a plan that assumes you have energy to cook from scratch nightly will fail. So:

- Frozen vegetables and frozen fish count. They are nutritionally fine and they do not spoil while you are having a bad week.
- Canned beans, canned fish, and pre-cooked grains count.
- Bagged salad counts.
- One cooking session on a better day can cover several worse days.
- A repeated meal you can make without thinking is worth more than variety you never execute.

Part two: the gut connection

This is where fibromyalgia diet work gets genuinely interesting, because the gut overlap is not speculation.

Irritable bowel syndrome and fibromyalgia overlap substantially, and research has documented this in both directions for decades (PMID 10606316). More recently, work comparing the gut bacterial communities of people with fibromyalgia against matched controls found meaningful differences in composition, with some of those differences correlating with symptom severity (PMID 31219947). A 2025 study went further, showing in mice that transferring gut bacteria from people with fibromyalgia produced pain and immune changes that paralleled the human condition, and that replacing that microbiota substantially reduced the pain response (PMID 40280127).

Two cautions about that last one. It is largely animal work, and the human component was a small open-label trial. Fecal transplantation is a research procedure, not something to attempt or to purchase. What it does justify is taking the gut seriously as part of your fibromyalgia picture rather than treating it as a separate complaint.

The low FODMAP approach

FODMAPs are fermentable carbohydrates found in foods including wheat, onion, garlic, certain fruits, some legumes, milk products containing lactose, and sugar alcohols. In sensitive people they draw water into the bowel and ferment rapidly, producing gas, bloating, pain, and irregular stools.

What the fibromyalgia research shows. In a four-week study of 38 women with fibromyalgia, a low FODMAP diet was followed by significant declines in pain scores, fibromyalgia severity, and fibromyalgia impact scores, along with roughly a 50% reduction in irritable bowel severity scores, and the improvement in fibromyalgia impact correlated with the improvement in gastrointestinal scores (PMID 28850525). In a randomized controlled trial, 46 women with fibromyalgia were assigned either to general healthy eating advice or to an anti-inflammatory diet excluding gluten, dairy, added sugar, and ultra-processed foods, with low FODMAP restriction in the first month. After three months, the intervention group improved significantly more on fibromyalgia impact, pain, pain inventory, fatigue, gastrointestinal symptoms, sleep quality, and quality of life. Notably, blood inflammatory markers did not change in either group (PMID 36091254).

A long-term follow-up study of irritable bowel patients with and without fibromyalgia found that an adapted, personalized version of the low FODMAP diet was sustainable over several years and that having fibromyalgia did not reduce adherence or the relief obtained (PMID 39408383).

How to do it without wrecking your diet. The single biggest mistake is treating low FODMAP as a permanent way of eating. It is not, and staying in the restriction phase long term narrows your nutrition and can hurt your gut bacteria.

1. **Restrict for two to six weeks.** Not longer.
2. **Reintroduce systematically.** One FODMAP group at a time, over two or three days, with a rest period before the next one.
3. **Personalize and expand.** You end at a personalized diet that includes everything you tolerate, which is usually far more than the restriction phase suggests (PMID 39408383).

Do this with a dietitian if you can. It is the part of this guide where professional help most clearly pays for itself.

Part three: the specific experiments

These are worth testing, one at a time, if the foundation is in place and you still have symptoms you want to chase. Each gets an honest strength-of-evidence label.

Gluten

Evidence strength: mixed, with the better-designed study being the more cautious one.

In a prospective study of 142 patients with fibromyalgia, the prevalence of non-celiac gluten sensitivity was 5.6%, which is similar to estimates in the general population. About 21.8% of patients responded to a six-week gluten-free diet through improvement in intestinal symptoms, and notably 74.2% of those responders did not meet the formal diagnostic criteria for gluten sensitivity. The authors concluded that a gluten-free diet cannot be systematically recommended for everyone with fibromyalgia, though it may be worth evaluating in those with diarrhea or specific intestinal findings (PMID 35900154).

A smaller study of 20 postmenopausal women with fibromyalgia and no celiac disease used a more telling design: six months gluten-free, then three months of unrestricted gluten, then gluten-free again. Widespread Pain Index and Symptom Severity scores improved on the gluten-free phases, rose again when gluten was reintroduced, and improved again on the rechallenge (PMID 37721353). That on-off-on pattern is more persuasive than a single before-and-after, and the study is still only 20 people with no control group.

How I read this. Gluten is not a universal fibromyalgia trigger, and the honest prior is that most people will not respond. It is a reasonable personal experiment if you have gut symptoms. **Get celiac testing before you remove it**, because testing after removal is unreliable.

How to test it fairly. Strict removal for four to six weeks, then a deliberate reintroduction for three days while you score symptoms daily. If nothing changes in either direction, you have your answer, and you get bread back.

Glutamate and aspartame

Evidence strength: genuinely conflicting, and I am going to show you both sides.

In one randomized study, 57 patients with fibromyalgia who also had irritable bowel syndrome followed a four-week diet excluding added glutamate and aspartame. Of the 37 who completed it, 84% reported that more than 30% of their symptoms resolved. Those responders were then randomized to a double-blind crossover challenge with monosodium glutamate or placebo. The glutamate challenge produced a significant return of symptoms and a worsening of fibromyalgia severity scores compared with placebo (PMID 22766026).

A different randomized study of 72 women with fibromyalgia compared eliminating monosodium glutamate and aspartame against a waiting list and found no significant difference in reported pain (PMID 23765203).

How I read this. The two studies asked slightly different questions. The first tested people who had already responded to elimination, which is a much more targeted question than the second study's broad elimination in everyone. Together they suggest that dietary glutamate may matter for a subset and probably not for the average person.

Practical version. This one rides along with reducing ultra-processed food, since that is where added glutamate and aspartame mostly live. You do not need a separate elimination protocol to test it. If you cut ultra-processed foods for four weeks and feel better, glutamate may be part of the reason, and a deliberate reintroduction will tell you more.

Weight, if it applies to you

Evidence strength: consistent association, limited intervention data.

A systematic review and meta-analysis of 41 articles found that obesity is common in fibromyalgia, with a pooled prevalence of 35.7%, and that it is associated with worse scores across pain severity, tender point count, stiffness, fatigue, physical functioning, sleep, cognition, and quality of life, although the correlations were weak on average and overall study quality was low. The review found only preliminary evidence that weight loss improves fibromyalgia symptoms (PMID 33676126). A pilot study of a 20-week behavioral weight loss program in overweight and obese women with fibromyalgia found an average loss of about 9 pounds, and that weight loss predicted reductions in fibromyalgia symptoms, pain interference, and quality of life measures (PMID 16253617).

How I want you to use this. Carefully. Weight is not a character issue and it is not the cause of your fibromyalgia. If your weight has been climbing and it bothers you, the data suggest modest loss may help symptoms modestly. If chasing weight loss makes you eat less than your body needs, worsens your relationship with food, or crowds out the foundation work, it is doing you more harm than good. Skip this section entirely if you have any history of disordered eating.

Part four: blood sugar and the afternoon crash

You know the pattern. A carbohydrate-heavy lunch, a hard energy drop by mid-afternoon, a craving for something sweet, and pain that feels louder by evening.

There is a research thread here worth knowing about. An observational study found an association between insulin resistance markers and fibromyalgia compared with control populations, independent of age, gender, and ethnicity, and the authors were explicit that this is a small cross-sectional study generating a hypothesis that needs randomized trials (PMID 33740353).

Important honesty note. An earlier and more widely publicized paper by the same lead researcher, suggesting insulin resistance might be the cause of fibromyalgia, was later retracted by the journal. I am telling you that because you will encounter that claim online presented as established fact, and because a guide that cites the retracted version to sound impressive is not a guide you should trust with anything else. Insulin resistance is an interesting association in fibromyalgia. It is not a proven cause.

What to actually do, which is sensible regardless of how that research resolves:

- **Anchor every meal with protein.** Eggs, fish, poultry, beans, tofu, plain yogurt if you tolerate dairy. This is the highest-yield habit in this section.
- **Pair carbohydrates with fat, fiber, or protein.** Fruit with nuts rather than fruit alone.
- **Stop drinking your sugar.** Sweetened drinks produce the sharpest swings.
- **Eat on a rhythm.** Long gaps followed by a large meal produce the biggest crashes. Three meals, or three meals with a small snack, beats skipping and then overcorrecting.
- **Do not skip breakfast to save calories** if your afternoons fall apart. That trade is usually a bad one for symptom stability.
- **A short walk after your largest meal** helps blunt the post-meal rise, and it doubles as the gentle movement that has the best evidence behind it in fibromyalgia overall (PMID 27377815).

Part five: eating for sleep

Sleep deserves its own section because in fibromyalgia it is not a side issue.

Sleep and pain run in both directions, and if anything the evidence that poor sleep worsens next-day pain is stronger than the reverse (PMID 24290442). A population study following people with chronic widespread pain over time found that reporting restorative sleep predicted the resolution of that pain, which is a hopeful finding: sleep is not just a symptom you suffer, it is a lever that changes the trajectory (PMID 18842606).

Food touches sleep in a few practical ways.

Caffeine. It has a long half-life, and in the afternoon it quietly costs you deep sleep even when you fall asleep fine. A reasonable experiment: no caffeine after noon for two weeks, and score your sleep. If you are drinking a lot, taper rather than quitting abruptly, because withdrawal headaches will muddy the result.

Alcohol. Here is a finding that surprises people. In a study of 946 patients with fibromyalgia, low and moderate drinkers reported lower symptom severity and better physical quality of life than nondrinkers (PMID 23497427). Read the design before you pour anything: this was self-reported, cross-sectional, and observational, meaning it cannot tell you whether alcohol helped, or whether people who feel better are simply more able to have a social drink. The authors said the reasons were unclear.

What I would actually tell you: this is not a reason to start drinking, and it is not a recommendation. Alcohol fragments the second half of the night, which is exactly the sleep you need most, and it interacts with many medications used in fibromyalgia, including sedatives, sleep aids, and opioids. If you drink, keep it low and not close to bedtime.

Evening eating. Large late meals, spicy food, and high-fat meals close to bed worsen reflux and fragment sleep for many people. Try to finish eating two to three hours before bed. If you truly need something, keep it small.

Overnight blood sugar. Some people wake at three in the morning and cannot get back down. If that is you, a small protein-containing snack in the evening is a harmless experiment worth two weeks of trial.

Hydration. Aim for pale yellow urine as your simple gauge. Being under-hydrated worsens fatigue, headache, and constipation, all of which you have enough of already. Front-load your fluids earlier in the day so you are not up at night, and if you take a diuretic or have a heart or kidney condition, ask your clinician what target is right for you rather than following a generic number.

Part six: a week you can actually cook

This is a scaffold, not a prescription. All generic foods, no products. Adjust freely for allergies and preferences, and if you are in a low FODMAP phase, swap onion and garlic for the green tops of scallions and for garlic-infused oil.

Breakfast, rotate three of these:

- Eggs with sauteed spinach and a slice of whole grain toast, or gluten-free toast if you are testing that
- Plain yogurt with berries, walnuts, and ground flaxseed (lactose-free yogurt if dairy bothers you)
- Oats cooked with milk of your choice, plus nut butter and blueberries
- Leftover dinner protein with fruit, which is a completely legitimate breakfast

Lunch, built the same way each time: a protein, a pile of vegetables, a fat, and a modest carbohydrate.

- Canned salmon or chickpeas over bagged greens, olive oil and lemon, with a small potato or rice
- Big salad with grilled chicken, cucumber, tomato, olives, pumpkin seeds, olive oil
- Lentil soup made in a batch, with a side salad
- Rice bowl with tofu or shrimp, shredded carrot, cabbage, sesame, and olive or sesame oil

Dinner, seven low-effort templates:

1. Baked salmon, roasted broccoli, roasted potatoes
2. Sheet pan chicken thighs with zucchini, peppers, and olive oil
3. Turkey and vegetable chili in a batch, freeze half
4. Stir-fried shrimp with mixed vegetables over rice
5. White fish baked in parchment with lemon and herbs, side of green beans
6. Bean and vegetable stew with olive oil and a green salad
7. Whatever you made in a batch this week, reheated, with a bagged salad on the side

Snacks: a handful of nuts, fruit with nut butter, hummus with cucumber and carrot, olives, hard-boiled eggs, plain yogurt.

Flare-day meals. Plan these before you need them, because you will not plan them during. Keep canned fish, frozen vegetables, frozen cooked rice, eggs, and pre-washed greens in the house. A can of sardines, frozen peas, and microwaved rice is a real meal and it takes four minutes.

A batch session on a good day: cook a grain, roast two trays of vegetables, cook a protein, mix a jar of olive oil vinaigrette. That is three or four assembled meals with no further cooking.

Running a fair diet experiment

The rules mirror the supplement guide, for the same reason: without them you cannot tell what worked.

1. **Baseline for two weeks.** Change nothing. Score pain out of ten, sleep out of ten, and gut symptoms out of ten most days. Count flare days.
2. **Change one thing.** The foundation first. Then, if you still want answers, one experiment at a time.
3. **Give it four to six weeks.** Not four days. Fibromyalgia varies week to week on its own, and short trials mostly measure noise.
4. **Reintroduce deliberately.** The reintroduction is the actual test. Improvement during restriction can be placebo, seasonal, or coincidence. A clean symptom return on reintroduction is real information.
5. **Write it down.** Memory reconstructs the story you already believe. Paper does not.
6. **Stop when it is not working.** A restriction that has done nothing after six weeks is costing you variety, nutrients, and joy for no return.

A hard boundary: if you find yourself with fewer than about 25 foods you feel safe eating, or if the rules are producing anxiety around meals, stop and talk to a clinician. Progressive food restriction is a known trap in chronic illness, and it is one of the few ways a diet guide can genuinely hurt you.

Realistic expectations

What is realistic. Modest, cumulative gains. Less bloating within two to three weeks of the gut work. Steadier afternoons within a couple of weeks of the blood sugar habits. A point off average pain or a couple fewer flare days per month over two to three months. Better sleep quality that then makes everything else easier. Those are the sizes of effect the studies in this guide actually measured.

What is not realistic. Remission from an eating pattern. Every serious review of this literature says the same thing: promising signals, small studies, poor quality, no single recommended diet (PMID 32878326, PMID 30735059). Anyone promising more than that is selling you something.

What tends to help most, in order. Fewer ultra-processed foods and more home cooking. A Mediterranean-style base. Gut-focused work if you have gut symptoms. Protein at every meal for blood sugar steadiness. Then the targeted experiments.

What food will not replace. Movement, which is the intervention with the strongest evidence in fibromyalgia (PMID 27377815). Sleep work. The medical care you are already receiving. Diet is a supporting player here, and it plays that role well.

When to seek professional care

Work with a clinician or registered dietitian if:

- You are planning an elimination diet and want it done in a way that does not cost you nutrition
- You have diabetes, kidney disease, liver disease, or take a blood thinner
- You are pregnant, nursing, or trying to conceive
- You have any history of disordered eating
- Your list of tolerated foods keeps shrinking
- You have gut symptoms that have never been properly evaluated

Seek medical care promptly for: unexplained weight loss, blood in the stool or black stools, difficulty or pain with swallowing, persistent vomiting, symptoms that wake you from sleep, new severe abdominal pain, or fever with abdominal pain. Those are not fibromyalgia symptoms and they are not diet-experiment territory.

Working with Root Health

This guide is general education by design, written so it is useful without knowing anything about your labs. Personalized work is different: it starts with testing, your history, your medication list, and the symptoms costing you the most, then sequences changes instead of piling them on.

If you want that, a Root Discovery Session is the right first step:

> ***Book a free Root Discovery Session: rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=fibromyalgia-diet***

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

> ***Get in touch: rootcauseunlocked.com/contact?utm_source=guide&utm_medium=ebook&utm_campaign=fibromyalgia-diet***

If you would rather start on your own, start tracking instead. Diet changes are exactly the kind of experiment that needs a written record, because the effects are gradual and memory is unreliable about gradual things:

> ***Free symptom tracker: rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=fibromyalgia-diet***

Read this alongside the companion **Fibromyalgia Supplement Guide**. Several nutrients discussed there work better inside the eating pattern described here, and the supplement guide carries the full medication interaction warnings that anyone considering supplements needs to read first, plus guidance on choosing quality products.

Reach me at (919) 662-0520 or at 503 US-70 East, Suite L, Garner, NC 27529.

References

All references below were verified against live PubMed records on 2026-08-02, including a check of publication type for retractions. Full verification detail, including what each citation does and does not support, is documented in `Fibromyalgia-Citation-Ledger-Supplements-Diet.md`.

1. Lowry E, Marley J, McVeigh JG, et al. Dietary Interventions in the Management of Fibromyalgia: A Systematic Review and Best-Evidence Synthesis. *Nutrients.* 2020;12(9):2664. PMID 32878326
2. Silva AR, Bernardo A, Costa J, et al. Dietary interventions in fibromyalgia: a systematic review. *Ann Med.* 2019;51(sup1):2-14. PMID 30735059
3. Metyas C, Aung TT, Cheung J, et al. Diet and Lifestyle Modifications for Fibromyalgia. *Curr Rheumatol Rev.* 2024;20(4):405-413. PMID 38279728
4. Rus A, Molina F, Ramos MM, et al. Extra Virgin Olive Oil Improves Oxidative Stress, Functional Capacity, and Health-Related Psychological Status in Patients With Fibromyalgia: A Preliminary Study. *Biol Res Nurs.* 2017;19(1):106-115. PMID 27443526
5. Casini I, Ladisa V, Clemente L, et al. A Personalized Mediterranean Diet Improves Pain and Quality of Life in Patients with Fibromyalgia. *Pain Ther.* 2024;13(3):609-620. PMID 38683449
6. Miller EE, Jayakrishnan S, Froio F, et al. Mediterranean diet and its low-antigenic and anti-inflammatory properties on fibromyalgia: a systematic review. *Clin Exp Rheumatol.* 2025;43(6):970-977. PMID 40576703
7. Rondanelli M, Faliva MA, Miccono A, et al. Food pyramid for subjects with chronic pain: foods and dietary constituents as anti-inflammatory and antioxidant agents. *Nutr Res Rev.* 2018;31(1):131-151. PMID 29679994
8. Martin Perez IM, Martin Perez SE. Assessment of ultra-processed food consumption and pro-inflammatory dietary load in adults diagnosed with fibromyalgia syndrome: A case-control study. *Nutr Health.* 2026;32(4):1243-1254. PMID 41021618
9. Sperber AD, Atzmon Y, Neumann L, et al. Fibromyalgia in the irritable bowel syndrome: studies of prevalence and clinical implications. *Am J Gastroenterol.* 1999;94(12):3541-3546. PMID 10606316
10. Minerbi A, Gonzalez E, Brereton NJB, et al. Altered microbiome composition in individuals with fibromyalgia. *Pain.* 2019;160(11):2589-2602. PMID 31219947
11. Cai W, Haddad M, Haddad R, et al. The gut microbiota promotes pain in fibromyalgia. *Neuron.* 2025;113(13):2161-2175.e13. PMID 40280127
12. Marum AP, Moreira C, Saraiva F, et al. A low fermentable oligo-di-mono saccharides and polyols (FODMAP) diet reduced pain and improved daily life in fibromyalgia patients. *Scand J Pain.* 2016;13:166-172. PMID 28850525
13. Silva AR, Bernardo A, de Mesquita MF, et al. An anti-inflammatory and low fermentable oligo, di, and monosaccharides and polyols diet improved patient reported outcomes in fibromyalgia: A randomized controlled trial. *Front Nutr.* 2022;9:856216. PMID 36091254
14. Lambiase C, Rossi A, Morganti R, et al. Adapted Low-FODMAP Diet in IBS Patients with and without Fibromyalgia: Long-Term Adherence and Outcomes. *Nutrients.* 2024;16(19):3419. PMID 39408383
15. Almirall M, Casellas F, Dot J, et al. Prevalence of non-coeliac gluten sensitivity and assessment of the response to gluten-free diet in a cohort of patients with fibromyalgia. *Rheumatology (Oxford).* 2023;62(3):1069-1077. PMID 35900154
16. Bruzzese V, Marrese C, Scolieri P, Pepe J. Efficacy of a gluten-free diet in reducing the widespread pain index and symptom severity scale in patients affected by fibromyalgia. *Reumatismo.* 2023;75(3). PMID 37721353

17. Holton KF, Taren DL, Thomson CA, et al. The effect of dietary glutamate on fibromyalgia and irritable bowel symptoms. *Clin Exp Rheumatol.* 2012;30(6 Suppl 74):10-17. PMID 22766026

18. Vellisca MY, Latorre JI. Monosodium glutamate and aspartame in perceived pain in fibromyalgia. *Rheumatol Int.* 2014;34(7):1011-1013. PMID 23765203

19. D'Onghia M, Ciaffi J, Lisi L, et al. Fibromyalgia and obesity: A comprehensive systematic review and meta-analysis. *Semin Arthritis Rheum.* 2021;51(2):409-424. PMID 33676126

20. Shapiro JR, Anderson DA, Danoff-Burg S. A pilot study of the effects of behavioral weight loss treatment on fibromyalgia symptoms. *J Psychosom Res.* 2005;59(5):275-282. PMID 16253617

21. Pappolla MA, Manchikanti L, Candido KD, et al. Insulin Resistance is Associated with Central Pain in Patients with Fibromyalgia. *Pain Physician.* 2021;24(2):175-184. PMID 33740353

22. Finan PH, Goodin BR, Smith MT. The association of sleep and pain: an update and a path forward. *J Pain.* 2013;14(12):1539-1552. PMID 24290442

23. Davies KA, Macfarlane GJ, Nicholl BI, et al. Restorative sleep predicts the resolution of chronic widespread pain: results from the EPIFUND study. *Rheumatology (Oxford).* 2008;47(12):1809-1813. PMID 18842606

24. Kim CH, Vincent A, Clauw DJ, et al. Association between alcohol consumption and symptom severity and quality of life in patients with fibromyalgia. *Arthritis Res Ther.* 2013;15(2):R42. PMID 23497427

25. Macfarlane GJ, Kronisch C, Dean LE, et al. EULAR revised recommendations for the management of fibromyalgia. *Ann Rheum Dis.* 2017;76(2):318-328. PMID 27377815

Dr. Adam Seay, DC · Root Health · Garner, NC · rootcauseunlocked.com

Educational content only. Not medical advice. Not a diagnosis. Not a substitute for your clinician. Talk with your prescriber or a registered dietitian before making major dietary changes.

Read this part first

This guide is educational. It is not medical advice, it does not diagnose fibromyalgia or anything else, and reading it does not make you my patient.

Some symptoms need evaluation before an exercise plan, not after one. New weakness, numbness that follows a nerve path, unexplained weight loss, fever, night pain that wakes you and will not settle, a new severe headache, chest pain or shortness of breath on exertion, or any sudden change in what your body does. Widespread pain is common. Those things are not, and they get looked at first.

If you have not had a medical evaluation for widespread pain, get one before you start here. Thyroid disease, anemia, vitamin D deficiency, inflammatory arthritis, sleep apnea and several other conditions produce a pain-and-fatigue picture that looks like fibromyalgia and is treated completely differently. The lab guide in this program covers what to ask for.

One condition-specific caution that has its own section below. If you experience post-exertional malaise, meaning a delayed and disproportionate crash a day or two after activity, the standard advice to increase steadily every week is not appropriate for you and has caused real harm in people with that pattern. Read the screening section in Part One before you start anything.

Why this guide exists

Look at what you have been given for fibromyalgia. There is a guide for your labs, a guide for your food, and a guide for your supplements.

The thing with the best evidence behind it did not have a guide at all.

That is not a small gap. Across the treatment options in this condition, movement is the one that the international guideline puts at the front, and it is the only one where the research is deep enough to argue about *dose* rather than about whether it works (PMID 27377815). Meanwhile the supplement guide in this program runs to thousands of words about compounds with far thinner support, and I say so in that guide too.

So this document is the missing piece. It is also the one I am most careful writing, because generic exercise advice does more damage in fibromyalgia than in any other condition I treat.

What you are actually buying, stated plainly

The evidence here is better than anywhere else in this program, and it is still not a promise.

What is solid: exercise beats no exercise in this condition, across multiple randomized trials and Cochrane reviews. That is about as settled as anything gets here.

What is genuinely uncertain: which *kind* of exercise is best. A 2025 blinded randomized trial compared progressive strength training, constant-intensity strength training, and walking. All three improved symptoms within their own groups. **There were no significant differences between them**, and no measure reached the threshold for a clinically important change (PMID 41674030). Read that honestly and it says two things at once. Doing something beats doing nothing. And arguing about which something is mostly arguing.

What the effect size actually is: modest. In the aerobic exercise dosing analysis, the pooled difference was small (PMID 39805734). Nobody is being cured here. People are getting function back, and function is what I would have you measure.

So I am not going to tell you movement fixes fibromyalgia. I am going to tell you it is the highest-yield thing on the list, that the dose is where people fail, and that this guide is mostly about the dose.

Part one: before you start

The physiology that explains why this feels unfair

Here is something worth knowing before your first session, because it validates an experience you have probably been told is in your head.

In healthy people, a bout of exercise produces short-term pain relief. It has a name, exercise-induced hypoalgesia, and it is one reason exercise feels good afterward.

In fibromyalgia, that response is absent. In a study comparing people with fibromyalgia, people with knee osteoarthritis, and pain-free controls, a five-minute submaximal isometric contraction produced exercise-induced hypoalgesia in the control group and produced **none** in either chronic pain group. Both chronic pain groups also showed lower vagal activity at rest (PMID 36711216). The same inconsistency shows up in a systematic review of isometric exercise across musculoskeletal pain populations, where the effect that appears reliably in healthy people did not appear reliably in people with pain (PMID 33601254).

Sit with what that means. When a healthy person exercises, their nervous system hands them a reward. When you exercise, it often does not. You are being asked to do the same work for none of the immediate payoff.

That is not a character problem and it is not deconditioning. It is a measured difference in how your nervous system responds to exertion.

And here is the part that makes it worth doing anyway. The benefit in fibromyalgia does not come from the session. It comes from the training effect across weeks. Impaired pain modulation has been shown to improve after exercise interventions rather than after single bouts (PMID 35399187, PMID 36660198), and the proposed mechanisms for why exercise helps in this condition are specifically about that longer adaptation (PMID 37484924).

So the deal is honest but harsh: you pay up front and you collect later. Which is exactly why the dose has to be set so the payments are affordable.

The screening question that changes everything

Do you experience post-exertional malaise?

That means a crash that arrives **delayed**, typically twelve to forty-eight hours after activity, that is **disproportionate** to what you did, and that involves a worsening of your whole symptom picture rather than just sore muscles.

Ordinary post-exercise soreness is not this. Soreness arrives within a day, sits in the muscles you used, and fades. Post-exertional malaise can arrive two days later, after a normal grocery trip, and take a week.

If the answer is yes, the standard advice does not apply to you. Programs built on steady weekly increases regardless of how you feel have caused documented harm in people with this pattern, which is why current review of that population treats pacing rather than graded escalation as the appropriate approach (PMID 41366804). The distinction is not academic and it is not a matter of attitude.

If the answer is yes: work from Part Three, the pacing model, and treat every progression rule in Part Two as optional rather than scheduled.

If the answer is no: you can use Part Two’s progression as written, still starting lower than you think you should.

If you are not sure, assume yes for the first month and find out. The cost of going too slowly is a slower start. The cost of going too fast is losing weeks.

Three honest expectations before your first session

You will not feel better afterward, at first. See the physiology above. Judge this at six to eight weeks, not at session three.

The first two weeks should feel absurdly easy. If your first session feels like a real workout, it was too much. This is the single most common failure and it is the reason people conclude exercise makes fibromyalgia worse.

Consistency beats intensity by a wide margin. In the dosing analysis, sessions as short as ten minutes appeared in the trials that worked, and useful interventions ran anywhere from three to twenty-four weeks (PMID 39805734). Ten repeatable minutes beats forty minutes you do twice and abandon.

Part two: the four kinds of movement, and what each one has behind it

Aerobic movement, the base of the whole thing

What it is. Anything that raises your breathing and heart rate and that you can keep going: walking, stationary cycling, swimming, pool walking, dancing to music.

The evidence. This is the most studied form in fibromyalgia and has its own Cochrane review (PMID 28636204), sitting inside a broader Cochrane overview of physical activity for chronic pain (PMID 28436583). A dedicated 2025 systematic review analyzed seventeen randomized trials specifically to work out the dose, using the frequency, intensity, type, time, volume and progression framework, and found in favor of aerobic exercise overall (PMID 39805734). An unsupervised walking program has its own randomized trial in women with fibromyalgia (PMID 32544346), which matters because it means this does not require a gym or a supervisor.

The dose, from that analysis. The trials that worked used frequencies from one to ten sessions per week, intensities from light to vigorous, sessions from ten to forty-five minutes, and programs from three to twenty-four weeks. Types included walking, stationary cycling, swimming, pool-based work, interval formats and music-based exercise.

What I take from that range. The successful trials are all over the map, which tells you the specific prescription matters less than finding one you will repeat. So: **start at ten minutes, three times a week, at an intensity where you could hold a conversation.** That is the floor, and the floor is deliberately embarrassing.

Progression, if you do not have post-exertional malaise. Add roughly ten percent to the duration only after two full weeks at the current amount with no increase in your symptom baseline. Duration first. Intensity much later, if at all.

Strength work, which does more than people expect

The evidence. A multi-center assessor-blinded randomized trial put 130 women with fibromyalgia through fifteen weeks of progressive resistance exercise, twice a week, against an active control. The resistance group improved significantly on knee extension strength, overall health status, current pain intensity, six-minute walk distance, elbow flexion strength, pain disability and pain acceptance (PMID 26084281).

Notice what is in that list. Not just strength. **Pain intensity, pain disability, and pain acceptance.** Getting stronger changed how much the pain cost, not only what the muscles could do.

One design detail from that trial is worth copying. The researchers used a person-centred model built specifically to support the participants' confidence about exercising, and they said plainly why: because activity-induced pain is a known risk in this condition. They designed around the flare rather than pretending it would not happen. That is the right posture and it is the one I use.

Stretching alone is not the same thing. A randomized trial comparing muscle stretching against resistance training found they are not interchangeable (PMID 29185675). Stretching has a place. It is not the strength program.

The dose. Twice a week. Start with body weight or the lightest band you own, one set of eight to ten repetitions per movement, five or six movements covering legs, pushing, pulling and trunk. Add repetitions before you add load. Take the first month to build the habit, not the strength.

And the honest counterweight. That 2025 blinded trial found progressive intensity did not beat constant moderate intensity or walking (PMID 41674030). So progress if it feels available. Do not treat progression as the point.

Water, which is where I start most people who are struggling

Why it is different. Buoyancy removes the load while keeping the movement, warmth helps the muscular guarding, and the water gives resistance in every direction without any equipment.

The evidence. Aquatic exercise in fibromyalgia has its own systematic review (PMID 38540665) plus a separate systematic review finding improvement specifically in self-reported sleep quality (PMID 37847348), which matters in a condition where sleep and pain feed each other. Pool-based work also appears among the effective aerobic formats in the dosing analysis (PMID 39805734), and comparative work on which therapeutic exercises help pain most in women with fibromyalgia includes aquatic modalities (PMID 40319533).

Who I put in the water first. Anyone whose land-based attempts have all ended in flares, anyone with significant joint pain layered on top of the fibromyalgia, and anyone who has become fearful of movement. Warm water lowers the stakes of the first attempt, and the first attempt is the one that decides whether there is a second.

Mind-body movement, and the trial that surprised people

The finding. A 52-week single-blind comparative effectiveness randomized trial in 226 adults with fibromyalgia compared classic Yang style tai chi against supervised aerobic exercise, the standard core treatment. Everyone improved. **The tai chi groups improved more** on the fibromyalgia impact questionnaire at 24 weeks, and the trial was published in the BMJ (PMID 29563100).

That is not a small alternative-medicine study. It is a large, long, properly designed trial in a major journal, and it found the gentle option beat the conventional one.

Why I think that is, though this part is my reasoning rather than the trial's finding. Tai chi combines low-intensity movement, balance work, breathing and attention, at a dose almost nobody flares on. In a condition where the main obstacle is that people cannot tolerate the dose, an intervention that is easy to tolerate has an enormous advantage that has nothing to do with the movements themselves.

What this changes practically. If you have failed at conventional exercise repeatedly, tai chi or qigong is not a consolation prize. It is the option with a head-to-head win.

What about education alongside the movement

Pain neuroscience education combined with resistance training has been studied in women with fibromyalgia (PMID 40686548), and low-intensity exercise has been shown to improve pain catastrophizing and other psychological measures (PMID 32455853). I mention this because understanding why your nervous system does what it does is not separate from the physical work. It changes how you interpret a hard day, and how you interpret a hard day determines whether you go back on Thursday.

Part three: pacing, which is the actual skill

Everything above is the menu. This is the method, and if you only take one thing from this guide, take this.

Pace by what is repeatable, not by what you can manage

The instinct is to set the dose by capacity: do what you can today. In fibromyalgia that produces the sawtooth. Good day, do everything, lose the week, recover, repeat, and after a year you are exactly where you started and you have concluded that exercise does not work for you.

The rule that breaks the cycle: find the amount you could do on a bad day, and do that amount on every day, including the good ones.

This is genuinely hard, and it is hard in a specific way. It requires you to stop on a good day while you still feel fine. That feels like leaving value on the table. It is not. It is the only way the baseline itself moves.

Pacing is a real technique with real research, including its complications

Activity pacing has been studied directly, and the literature is more interesting than the slogan version. One study found pacing associated with **both better and worse** symptoms depending on how it was being used (PMID 27322396), and related work on activity management patterns found that the pattern matters more than the label (PMID 28591081). Patients themselves have been asked what they think of pacing and of structured pacing frameworks (PMID 26385155).

The distinction those studies point at is this. **Pacing used to expand what you do is helpful. Pacing used to justify doing less and less is not.** The same word covers both, which is why the direction of travel matters more than the technique.

So the test I would apply monthly: is my baseline going up, holding, or quietly shrinking. If it is shrinking, this has stopped being pacing.

The practical version

Set a floor. Pick the amount you are confident you could do on a bad day. Halve it. That is week one.

Hold it for two weeks. Every scheduled day, including good ones, including days you feel capable of much more.

Then evaluate on the baseline, not on the sessions. Did your general symptom level get worse, stay the same, or improve. Only if it stayed the same or improved do you increase.

Increase by about ten percent, duration first.

When a flare happens, and it will: drop to half your current amount rather than to zero. Going to zero is what turns a three-day flare into a three-week restart. Hold the half amount until the flare settles, then return to where you were, not to where you were heading.

Build in rest days deliberately rather than taking them when forced.

Part four: how to know it is working

Pain is a poor instrument here. It moves for reasons that have nothing to do with your program, and in a condition this variable, judging by pain alone will mislead you in both directions.

Track function instead. Four things, once a week, thirty seconds:

How far can I walk without paying for it tomorrow. One number, in minutes or blocks.

How many days this week did I do my planned movement. Adherence, not intensity. This is the number that predicts everything else.

How many crash days did I have. Days where the whole picture worsened and normal activity stopped.

One thing I did this week that I could not do last month. Written in words, not numbers. Carried the laundry upstairs. Stood through a whole conversation. Went to the thing and stayed.

The tracker at rootcauseunlocked.com/tracker is free and does this for you. In this condition it earns its place, because the changes are gradual and memory across weeks is unreliable in exactly the direction that makes people quit.

Judge the program at eight weeks, not at two. And judge it on those four numbers, not on how any single session felt.

When to stop and get evaluated instead

- New weakness, numbness in a nerve pattern, or loss of bladder or bowel control. Same day.
- Chest pain, unusual shortness of breath, or fainting with exertion. Same day.
- A flare that has not settled after two weeks at half dose.
- Steadily shrinking capacity over two months despite honest pacing. That pattern deserves a fresh look at the diagnosis rather than a harder program.
- Post-exertional malaise that is worsening rather than stabilizing.
- Any new symptom that does not fit the pattern you know.

Working with Root Health

This guide is general education by design, written so it is useful without knowing anything about your case. Personalized work is different: it starts with your history, your other conditions, what you have already tried and how it went, and it sequences the changes instead of stacking them.

If you want that, a Root Discovery Session is free and is the right first step. Book at rootcauseunlocked.com/discovery.

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

If you would rather start on your own, the symptom tracker at rootcauseunlocked.com/tracker is free and works anywhere.

Reach me at (919) 662-0520 or at 503 US-70 East, Suite L, Garner, NC 27529.

This guide is educational. It is not medical advice, diagnosis, or treatment, and reading it does not create a doctor and patient relationship.

References

1. Macfarlane GJ, Kronisch C, Dean LE, et al. EULAR revised recommendations for the management of fibromyalgia. *Ann Rheum Dis*. 2017. PMID 27377815

2. Bidonde J, Busch AJ, Schachter CL, et al. Aerobic exercise training for adults with fibromyalgia. *Cochrane Database Syst Rev*. 2017. PMID 28636204

3. Geneen LJ, Moore RA, Clarke C, et al. Physical activity and exercise for chronic pain in adults: an overview of Cochrane reviews. *Cochrane Database Syst Rev*. 2017. PMID 28436583

4. Casanova-Rodriguez D, Ranchal-Sanchez A, Rodriguez RB, Jurado-Castro JM. Aerobic exercise prescription for pain reduction in fibromyalgia: a systematic review and meta-analysis. *Eur J Pain*. 2025. PMID 39805734 (see published erratum, *Eur J Pain* 2025;29(9):e70105)

5. Wang C, Schmid CH, Fielding RA, et al. Effect of tai chi versus aerobic exercise for fibromyalgia: comparative effectiveness randomized controlled trial. *BMJ*. 2018. PMID 29563100

6. Larsson A, Palstam A, Lofgren M, et al. Resistance exercise improves muscle strength, health status and pain intensity in fibromyalgia: a randomized controlled trial. *Arthritis Res Ther*. 2015. PMID 26084281

7. Pontes-Silva A, Dibai-Filho AV, Melo TS, et al. Strength training versus walking on the fibromyalgia impact: a blinded randomised controlled trial. *Musculoskeletal Care*. 2025. PMID 41674030

8. Aquatic exercise in physical therapy treatment for fibromyalgia: systematic review. *Healthcare (Basel)*. 2024. PMID 38540665

9. Aquatic therapy improves self-reported sleep quality in fibromyalgia patients: a systematic review. *Sleep Breath*. 2024. PMID 37847348

10. Bossenger NR, Lewis GN, Rice DA, Shepherd D. The autonomic and nociceptive response to acute exercise is impaired in people with knee osteoarthritis. *Neurobiol Pain*. 2023. PMID 36711216

11. Bonello C, Girdwood M, De Souza K, et al. Does isometric exercise result in exercise induced hypoalgesia in people with local musculoskeletal pain? A systematic review. *Phys Ther Sport*. 2021. PMID 33601254

12. Impaired conditioned pain modulation was restored after a single exercise session in individuals with and without fibromyalgia. *Pain Rep*. 2022. PMID 35399187

13. The effects of a 15-week physical exercise intervention on pain modulation in fibromyalgia. *Neurobiol Pain*. 2023. PMID 36660198

14. Mechanisms hypothesized for pain-relieving effects of exercise in fibromyalgia: a scoping review. *Ther Adv Musculoskelet Dis*. 2023. PMID 37484924

15. Promoting unsupervised walking in women with fibromyalgia: a randomized controlled trial. *Psychol Health Med*. 2021. PMID 32544346

16. Low-intensity physical exercise improves pain catastrophizing and other psychological and physical aspects in women with fibromyalgia. *Int J Environ Res Public Health*. 2020. PMID 32455853

17. Muscle stretching exercises and resistance training in fibromyalgia: which is better? A three-arm randomized controlled trial. *Eur J Phys Rehabil Med*. 2018. PMID 29185675

18. The most effective therapeutic exercises for pain intensity in women with fibromyalgia: a systematic review with network meta-analysis. *Braz J Phys Ther*. 2025. PMID 40319533

19. Pain neuroscience education and resistance training in women with fibromyalgia. *Pain Res Manag*. 2025. PMID 40686548

20. Activity pacing is associated with better and worse symptoms for patients with long-term conditions. *Clin J Pain*. 2017. PMID 27322396

21. Pain-related activity management patterns and function in patients with fibromyalgia syndrome. *Clin J Pain*. 2018. PMID 28591081

22. Exploring patients' opinions of activity pacing and a new activity pacing questionnaire for chronic pain and fatigue. *Physiotherapy*. 2016. PMID 26385155

23. Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS): diagnosis and management. *J Transl Med*. 2025. PMID 41366804

Please read this first

This guide is educational. It is not medical advice, it does not diagnose anything, and it does not replace your own clinician. Nothing here is a treatment for fibromyalgia, and nothing here is a substitute for the care you are already receiving.

Talk with your prescriber before you start, stop, or change any supplement. *Fibromyalgia is commonly managed with medications (duloxetine, milnacipran, amitriptyline, pregabalin, gabapentin, cyclobenzaprine, tramadol, SSRIs) that interact with several of the ingredients discussed in these pages. Some of those interactions are serious. The safety sections are not filler. They are the most important part of this guide.*

If you are pregnant, nursing, trying to conceive, under 18, have kidney or liver disease, a bleeding disorder, or take a blood thinner, treat every entry here as "ask first" rather than "try it."

Supplements in the United States are regulated as food, not as drugs. Quality varies. Choose products that carry independent third-party testing, and bring the actual label to your appointments.

What you are actually buying back

Nobody wakes up wanting a supplement routine. What people with fibromyalgia want back is specific: a night of sleep that actually restores something. A morning that does not start at a four out of ten before your feet hit the floor. A week with fewer flare days than good days. Enough steady energy to finish what you started.

That is the frame for this entire guide. Every entry gets judged against one question: does it help you get a piece of your life back? If the honest answer for a given ingredient is “the evidence is thin,” you will read that here in plain language instead of a sales pitch.

Supplements are supportive, not magic

Here is the part most supplement guides skip.

Fibromyalgia involves the way your nervous system processes pain signals, and it usually travels with three companions: unrefreshing sleep, fatigue, and gut symptoms. No capsule reprograms a sensitized nervous system. The interventions with the strongest evidence base in fibromyalgia are not supplements at all. European rheumatology guidelines put graded exercise at the top of the list, with education, sleep work, and psychological and physical therapies close behind, and they rate most other options far lower (PMID 27377815).

I use supplements in practice, and I think they earn their place. But they earn it as support: correcting a real nutrient shortfall, taking pressure off sleep, giving mitochondria better raw material, calming a gut that is feeding into the whole picture. They are the assist, not the play.

So the honest sequence is:

1. **Movement you can actually sustain.** Even small amounts, built slowly, done consistently.
2. **Sleep as a project**, not an afterthought.
3. **Food that steadies you** (see the companion Fibromyalgia Diet Guide).
4. **Then targeted supplements**, ideally guided by actual lab values.
5. **Alongside whatever medical care you are already receiving** (PMID 38927473).

If you skip the first three and buy the fourth, you will spend money and feel unchanged. I would rather tell you that now than let you learn it over six months.

How to read this guide

The tiers. Foundational means "most likely to matter for most people with fibromyalgia, and easiest to get right." Targeted means "pick based on which symptom is running your life." Ask-first means "the interaction risk is real enough that this belongs in a conversation with your prescriber before you order anything."

Typical studied range. Where a dose appears, it is the range used in the published research being cited, presented so you can discuss it intelligently with your clinician. It is not a prescription for you. Your kidneys, your liver, your medications, your labs, and your body weight all change the right answer.

The claims language is deliberate. You will read "supports," "may help," and "is being studied for." You will not read "treats," "fixes," or "reverses," because those words would be false, and because they are how people end up spending hundreds of dollars on hope.

One change at a time. Start one product, hold everything else steady, give it six to eight weeks, and write down what happened. Starting five things at once guarantees you will never know which one earned its place, and it makes any side effect impossible to trace.

Track something simple. Two numbers, most days: pain out of ten, and sleep quality out of ten. Add a flare-day tally per week. Twelve weeks of that beats any amount of "I think it might be helping."

Why the evidence for nutrients looks thinner than it is. A large outcome trial costs tens to hundreds of millions of dollars, and that money only exists when a patent can pay it back. Nobody can patent magnesium, so the hierarchy this guide grades against systematically over-documents patentable drugs and under-documents nearly everything else. A Cochrane methodology review quantified the tilt: manufacturer sponsorship produces more favourable results and conclusions than independent funding, through a bias that standard quality checks do not catch (PMID 28207928). So when this guide says "no evidence," it will tell you which of three things it means: *tested and failed* (a real trial ran and the answer was no), *never properly tested* (nobody could afford the trial, so absence of evidence is not evidence of absence), or *tested small and won* (real but modest trials, usually on markers rather than outcomes, because markers are what a small budget can afford).

The same skepticism runs in both directions. The asymmetry earns nutrients humility, not automatic credit: when isolated nutrients have received large public trials the results were mixed, and small nutrient trials carry their own publication bias toward positive findings, just as drug trials tilt toward their sponsor. This guide names the funding and the evidence tier on both sides and lets you see them.

Before you start: the medication conversation

Fibromyalgia care very often includes serotonin-active medication. That single fact drives most of the safety content in this guide.

Serotonin-active prescriptions commonly used in fibromyalgia: duloxetine, milnacipran, venlafaxine, amitriptyline and other tricyclics, SSRIs (fluoxetine, sertraline, escitalopram, paroxetine, citalopram), tramadol, and triptans for migraine. Cyclobenzaprine is structurally related to tricyclics and belongs on this list too.

Supplements that add serotonin activity on top of those: 5-HTP, SAMe, St John's wort, L-tryptophan, and some combination "mood" or "sleep" blends that contain them without saying so on the front of the label.

Combining serotonin-active drugs with serotonin-active supplements can cause serotonin syndrome, which ranges from agitation, sweating, tremor, and diarrhea to a medical emergency with high fever and muscle rigidity (PMID 37309284). This is the single most important safety point in this document. If you take any medication on that first list, the ingredients on the second list are prescriber conversations, not shopping-cart decisions.

Other interaction categories worth naming:

- **Sedating medications** (pregabalin, gabapentin, cyclobenzaprine, benzodiazepines, sleep aids, opioids) stack with sedating supplements such as melatonin. More grogginess, more fall risk.
- **Blood thinners and antiplatelet drugs** (warfarin, apixaban, clopidogrel, daily aspirin) stack with higher-dose fish oil and with several botanicals. Bleeding risk is the concern.
- **St John's wort** speeds up the liver's clearance of a long list of drugs, including oral contraceptives, some blood thinners, transplant medications, and several antidepressants, and can cause serious loss of effect (PMID 31742659). Do not self-start it.
- **Kidney or liver disease** changes the safety of magnesium, creatine, and fat-soluble vitamins.
- **Pregnancy and nursing:** almost nothing in this guide has been studied for safety in pregnancy. Default to "not without your obstetric provider's sign-off."

Bring the label. Not the marketing name, the actual ingredient panel. Half of the interaction problems I see come from blends where the risky ingredient is fourth on the list.

Tier 1: The foundational few

These are the ones with either the most consistent fibromyalgia-specific research or the most plausible chance of correcting something genuinely low. Start here, and ideally start with labs.

Vitamin D

What it is. A fat-soluble vitamin your skin makes from sunlight and your liver and kidneys activate. Low levels are common in people with chronic pain and in people who spend most of their time indoors, which describes a lot of fibromyalgia life.

Why it may help. Vitamin D deficiency shows up frequently in fibromyalgia populations, and reviews have looked at whether it tracks with symptom severity (PMID 35071984). In a small randomized placebo-controlled trial of women with fibromyalgia and low vitamin D, bringing blood levels up over 20 weeks was associated with a meaningful reduction in pain scores compared with placebo (PMID 24438771). A 2022 meta-analysis of five randomized studies covering 315 participants found that vitamin D supplementation improved Fibromyalgia Impact Questionnaire scores, though it did not find a statistically significant difference on the pain visual analogue scale (PMID 35596576). Read that split carefully: overall symptom impact improved, the pain-only measure did not reach significance, and the studies were small.

Typical studied range. The trials were designed to move a low blood level into the normal range rather than to test one fixed dose, which is exactly why the blood test comes first (PMID 24438771). Ask your clinician to check 25-hydroxy vitamin D and to set the dose from the result. Retest rather than assuming.

Timing. With a meal containing fat, which improves absorption.

Cautions. Vitamin D is fat-soluble and accumulates. High doses over time can raise blood calcium, which is not benign. It interacts with thiazide diuretics, some seizure medications, and digoxin. This is a test-and-dose nutrient, not a “more must be better” nutrient.

Magnesium

What it is. A mineral involved in hundreds of enzyme reactions, including muscle relaxation, nerve signaling, and energy production. Common forms are glycinate (gentler on the gut, often chosen for evening use), citrate (more likely to loosen stool), malate, and oxide (poorly absorbed, mostly a laxative).

Why it may help. A controlled study in 60 premenopausal women with fibromyalgia found that blood and red-cell magnesium levels were lower than in healthy controls, and that eight weeks of magnesium citrate at 300 mg daily reduced tender point counts, tender point index, Fibromyalgia Impact Questionnaire scores, and depression scores, with broader improvement when it was combined with low-dose amitriptyline (PMID 22271372). A literature review of magnesium in fibromyalgia concluded that the rationale is reasonable and the human evidence is still limited (PMID 34392734). A separate evidence summary looking specifically at magnesium plus malic acid found the certainty of evidence low and the effect uncertain (PMID 31150373). So: worth trying, honestly described as promising rather than proven.

Typical studied range. 300 mg daily of elemental magnesium as magnesium citrate for eight weeks in the trial above (PMID 22271372).

Timing. Evening tends to suit people best, especially glycinate, because of the muscle and sleep angle.

Cautions. Loose stools are the usual limiting side effect, and citrate is the worst offender for that. Magnesium can interfere with the absorption of some antibiotics, thyroid medication, and bisphosphonates, so separate them by several hours. **If you have reduced kidney function, do not take magnesium without your clinician's approval.** Impaired kidneys cannot clear it well.

Omega-3 fatty acids (EPA and DHA)

What it is. The long-chain fats found in oily fish, with algae-derived versions available for people who do not eat fish.

Why it may help. In a randomized, double-blind, placebo-controlled trial of 120 patients with fibromyalgia, eight weeks of a high oral dose of omega-3 was associated with significant improvement in the Widespread Pain Index, Symptom Severity Scale, pain visual analogue scale, and Fibromyalgia Impact Questionnaire compared with placebo (PMID 39377412). That is a single trial, and it deserves confirmation, but it is a reasonably sized one with patient-reported outcomes that matter. Broader nutrition reviews of chronic pain also place omega-3 intake inside an anti-inflammatory dietary pattern rather than treating it as a standalone remedy (PMID 29679994).

Typical studied range. The fibromyalgia trial tested a high dose over eight weeks and did not compare lower doses (PMID 39377412). Across pain and inflammation research generally, the range studied most often is 1 to 3 grams per day of combined EPA plus DHA. Note that this refers to actual EPA plus DHA content, not total “fish oil” on the front of the bottle, which is usually a much bigger number.

Timing. With meals. Splitting the dose reduces fishy repeat.

Cautions. Higher doses can increase bleeding tendency, which matters if you take a blood thinner, an antiplatelet drug, or daily aspirin, or if you have surgery scheduled. Ask your surgeon about stopping in advance. Fish allergy is an obvious exclusion, and algae-based products are the workaround.

Tier 2: Targeted support

Pick by the symptom that is costing you the most, not by the length of the list.

If sleep is the problem

Unrefreshing sleep is not a side issue in fibromyalgia. It sits close to the center, and it feeds pain the next day. Anything that genuinely improves sleep tends to pay you back twice.

Melatonin

What it is. A hormone your brain releases as darkness falls. It is a timing signal more than a sedative, which is why the timing of the dose matters as much as the amount.

Why it may help. In a double-blind, placebo-controlled study of 101 patients with fibromyalgia, melatonin at 3 mg or 5 mg nightly, used alone or alongside fluoxetine for eight weeks, was associated with significant reductions in Fibromyalgia Impact Questionnaire scores (PMID 21158908). A systematic review pooling four studies covering 98 patients found that all of them reported positive effects of melatonin on fibromyalgia symptoms with no major adverse events, while also flagging significant heterogeneity and the need for better trials (PMID 31783341). Small literature, consistent direction, low risk profile.

Typical studied range. 3 mg to 5 mg at night in the fibromyalgia study above (PMID 21158908). More is not better with melatonin. Many people do better on less.

Timing. Thirty to sixty minutes before bed, in a dim room. Scrolling in a bright screen at the same time cancels most of the point.

Cautions. Next-morning grogginess is the common complaint and usually means the dose is too high. It adds to the sedation of pregabalin, gabapentin, cyclobenzaprine, benzodiazepines, and opioids. Caution with blood thinners, with immune-suppressing medication, and in seizure disorders. Not established as safe in pregnancy or nursing. It can affect blood sugar and blood pressure control in some people.

Magnesium glycinate at night

Covered in Tier 1. Many people find the evening dose does double duty for muscle discomfort and sleep onset. The fibromyalgia trial data sit with magnesium citrate (PMID 22271372); choosing glycinate is a tolerability decision, not a stronger-evidence decision, and I want you to know the difference.

If pain is the problem

Palmitoylethanolamide (PEA), usually paired with acetyl-L-carnitine

What it is. PEA is a fatty acid compound your own body produces that acts on inflammation and pain signaling pathways. Acetyl-L-carnitine is a form of carnitine involved in moving fats into mitochondria for energy, with additional effects on nerve function.

Why it may help. In a randomized controlled study, patients already on stable duloxetine and pregabalin were randomized either to continue that alone or to add PEA and acetyl-L-carnitine. Over 24 weeks, the group that added the two supplements showed additional significant improvement in the Widespread Pain Index, the revised Fibromyalgia Impact Questionnaire, and the modified Fibromyalgia Assessment Status score (PMID 37378482). This is one of the more practically useful studies in the whole supplement literature for fibromyalgia, because it tested exactly the situation most readers are in: already on medication, still symptomatic, wondering whether adding something helps.

Typical studied range. PEA 600 mg twice daily and acetyl-L-carnitine 500 mg twice daily, as an addition to existing prescriptions (PMID 37378482).

Timing. Split morning and evening, as studied.

Cautions. This was an add-on study, not a replacement study. Nothing here suggests reducing a prescription, and you should not change one without your prescriber. Acetyl-L-carnitine can be stimulating for some people, so favor earlier dosing if it disturbs your sleep. Carnitine is generally avoided in people with seizure disorders without medical guidance, and people with thyroid conditions should ask, since carnitine can interact with thyroid hormone action.

Acetyl-L-carnitine on its own

Why it may help. A double-blind multicenter trial randomized 102 patients with fibromyalgia to acetyl-L-carnitine or placebo. Both groups improved early. By week ten, the placebo group had plateaued while the carnitine group kept improving, producing statistically significant between-group differences in total myalgic score, tender point count, depression, musculoskeletal pain, and most quality of life measures (PMID 17543140).

Typical studied range. 1,500 mg daily by mouth in the later phase of that trial. Note honestly that the first two weeks of the study also included intramuscular injections, so the oral-only effect is less certain than the headline suggests (PMID 17543140).

Cautions. As above. Ask before combining with thyroid medication or if you have a seizure history.

If fatigue and post-exertion crashes are the problem

Coenzyme Q10

What it is. A compound your mitochondria use in the electron transport chain, meaning it sits directly in the machinery that turns food into usable energy. Sold as ubiquinone or as ubiquinol.

Why it may help. A randomized, double-blind, placebo-controlled trial of 20 patients with fibromyalgia found that 300 mg daily of CoQ10 for 40 days was associated with reduced Fibromyalgia Impact Questionnaire scores, with the most prominent reductions in the pain, fatigue, and morning tiredness subscales, alongside changes in oxidative stress and mitochondrial gene expression markers (PMID 23458405). A more recent randomized crossover trial testing a combination of CoQ10, tryptophan, and magnesium found something more mixed: the primary endpoint of fatigue did not improve significantly more than placebo, while pain, sleep quality, and total symptom impact did improve on the supplement, with good tolerability (PMID 40151031).

How to read that honestly. The most enthusiastic CoQ10 fibromyalgia results come from a small number of very small studies from a single research group, and the largest recent independent trial missed its own primary endpoint. CoQ10 is a reasonable thing to try if fatigue is your main complaint. It is not the breakthrough that supplement marketing makes it out to be.

Typical studied range. 300 mg daily, taken in divided doses, in the trial above (PMID 23458405).

Timing. With a fat-containing meal, since absorption depends on it. Earlier in the day is usually better, as some people find it activating.

Cautions. CoQ10 can reduce the effect of warfarin, so anyone on it needs prescriber involvement and monitoring. It may modestly lower blood pressure and blood sugar, which matters if you are treated for either. Not established in pregnancy or nursing.

Creatine monohydrate

What it is. A compound stored in muscle that recycles quick-burst energy. Not just for athletes.

Why it may help. A 16-week randomized, double-blind, placebo-controlled trial in fibromyalgia found that creatine raised muscle phosphorylcreatine content and improved lower-body and upper-body muscle strength compared with placebo, with no serious adverse events. Be clear about the limits: that same trial found no meaningful changes in pain, cognition, sleep quality, or quality of life (PMID 23554283).

So why include it. Because strength is not a vanity outcome in fibromyalgia. If graded movement is the intervention with the best evidence behind it, then something that makes the movement possible without wrecking you afterward supports the thing that actually works. Just buy it for the right reason: strength and exercise capacity, not pain relief.

Typical studied range. Creatine monohydrate is most commonly used at about 3 to 5 grams daily for maintenance in adults.

Timing. Any time of day, with fluid. Consistency matters more than timing.

Cautions. Requires good hydration. Ask your clinician first if you have any kidney condition or take medications that affect the kidneys. Some people gain a small amount of water weight in the first weeks, which is expected and is not fat gain.

If your gut is part of the picture

Gut symptoms and fibromyalgia travel together often enough that ignoring the gut leaves results on the table. Research has found that the gut bacterial community in people with fibromyalgia differs from that of matched controls, with some of those differences correlating with symptom severity (PMID 31219947).

Probiotics and prebiotic fiber

Why they may help. In a randomized, double-blind, placebo-controlled trial in women with fibromyalgia, eight weeks of a probiotic was associated with reduced pain scores versus placebo, along with improvements from baseline in sleep quality, depression, and anxiety measures, while the prebiotic inulin arm improved sleep quality and pain scores (PMID 37224267). A separate pilot randomized controlled trial found that a multispecies probiotic improved measures of impulsivity and decision-making, which is a small but real signal on the cognitive side of fibromyalgia (PMID 30026567).

The honest counterweight. A larger randomized, double-blind, placebo-controlled trial of a multi-strain probiotic for the gastrointestinal symptoms of fibromyalgia found no benefit over placebo on its primary outcome or on any secondary outcome (PMID 34681287). Different strains, different targets, different results. Probiotics are not one product, and the honest summary is "mixed, possibly helpful, strain-specific, and not yet predictable."

Typical studied range. In the positive trial, a probiotic providing 4 x 10^10 colony forming units per day, or 10 grams per day of inulin, over eight weeks (PMID 37224267).

Timing. Consistent daily use. Follow the label for with or without food, since it depends on the strain and the formulation.

Cautions. Prebiotic fiber, including inulin, commonly increases gas and bloating at first and can be genuinely uncomfortable if you also have irritable bowel syndrome. Start low. People who are significantly immune-compromised, have a central line, or are critically ill should not start probiotics without medical guidance. If you are following a low FODMAP approach (see the Diet Guide), be aware that inulin is high FODMAP.

Tier 3: Ask your practitioner first

Everything in this tier has published fibromyalgia data behind it. Every one of them also has an interaction profile serious enough that I do not want you starting it from a guide, including this one. These are worth discussing. They are not worth improvising.

5-HTP (5-hydroxytryptophan)

What it is. A direct precursor to serotonin, one metabolic step closer than tryptophan.

What the research shows. A double-blind, placebo-controlled study of 50 patients with primary fibromyalgia reported significant improvement across the clinical parameters studied, with mild and transient side effects, and the authors themselves called for further controlled studies to define its value (PMID 2193835). That study is from 1990, it is small, and the literature has not meaningfully built on it since.

Why it is in this tier and not the last one. Because a serotonin precursor added to a serotonin-active prescription is the textbook setup for serotonin syndrome, which can escalate from tremor and agitation to a life-threatening emergency (PMID 37309284). Duloxetine, milnacipran, SSRIs, tricyclics, tramadol, and triptans are all common in fibromyalgia care. This is not a theoretical concern.

Dose. Deliberately not listed here. If you and your prescriber decide this is worth trying, they should set the dose, know your full medication list, and tell you exactly which symptoms mean “stop and call.”

Cautions. Do not combine with any serotonergic medication without prescriber supervision. Not for use in pregnancy or nursing. Stop before surgery and tell your anesthesia team, because several anesthetic and pain agents are serotonergic.

SAMe (S-adenosylmethionine)

What it is. A compound involved in methylation reactions throughout the body, studied for both mood and pain.

What the research shows. A six-week double-blind trial of 44 patients with primary fibromyalgia found that 800 mg daily of oral SAMe improved clinical disease activity, pain over the previous week, fatigue, morning stiffness, and mood compared with placebo, while tender point score and muscle strength did not differ (PMID 1925418). A later double-blind study in secondary fibromyalgia did not find the same benefit (PMID 9543578). Mixed, with the positive study being the older and better known of the two.

Typical studied range. 800 mg daily by mouth for six weeks (PMID 1925418).

Cautions. SAMe is serotonin-active. The same serotonin syndrome concern applies as with 5-HTP (PMID 37309284). It can also trigger agitation or mood elevation in people with bipolar disorder, so a personal or family history of bipolar illness is a specific reason to avoid self-starting it. Not established in pregnancy or nursing.

St John's wort

Included here so you know to avoid self-starting it, not because I am recommending it. It has no convincing fibromyalgia-specific evidence base, it is serotonin-active, and it induces liver enzymes and drug transporters in a way that reduces blood levels of a long list of medications, including oral contraceptives, some anticoagulants, immunosuppressants, and several antidepressants (PMID 31742659). If you are already taking it, tell every prescriber you see. Please do not add it on your own.

Iron, only if your labs say so

Low iron stores can cause fatigue that looks exactly like the fatigue you are already fighting, so this is worth testing rather than guessing. In a placebo-controlled trial of intravenous iron in iron-deficient patients with fibromyalgia, the primary endpoint was not met, while several secondary measures including Fibromyalgia Impact Questionnaire total score, pain inventory score, and fatigue score improved more than placebo (PMID 29149437). Note that this was intravenous iron given under medical supervision, which is not the same thing as a bottle of iron tablets.

Ask for ferritin plus a full iron panel and a complete blood count. If your iron stores are truly low, correcting that is a medical decision with a real dose and real follow-up testing. Unnecessary iron supplementation is not harmless: it commonly causes constipation and nausea, and iron overload is a genuine risk in people who do not need it. Iron is also a leading cause of poisoning deaths in young children, so if you keep it in the house, keep it locked away.

What I deliberately left out, and why

You have seen these on fibromyalgia supplement lists. Here is why they are not on mine.

- **NAD precursors (NMN, NR).** Interesting biology, no meaningful fibromyalgia trial evidence, expensive. If that changes, I will change this guide.
- **High-dose “detox” and chelation products.** Chelating agents and provoked heavy metal testing are not appropriate self-directed interventions, and marketed detox drops do not have the fibromyalgia evidence to justify what they cost or the risk of steering you away from care that does work.
- **D-ribose.** The most-cited fibromyalgia work here is a small open-label pilot with no placebo group, which is not enough to earn a recommendation.
- **Curcumin, boswellia, and similar anti-inflammatory botanicals.** They have research in other pain conditions, but fibromyalgia-specific human evidence is lacking, and curcumin products commonly interact with blood thinners. If you want to include turmeric, food is a reasonable place to do it.
- **Very large multi-ingredient stacks.** If you start eight products at once, you cannot tell what is working, you cannot trace a side effect, and you have quietly multiplied your interaction risk. The economics are not in your favor either.

Telling you what to skip is part of what you paid for.

Running a fair 12-week trial

This is how you find out what actually works for your body instead of collecting bottles.

Weeks 1 and 2, baseline. Change nothing. Score pain out of ten and sleep out of ten most days. Count flare days. Write down your current medications and supplements, doses included.

Week 3, add one thing. Choose the tier 1 item that fits your labs and your biggest symptom. Start at the low end of the studied range. Keep scoring.

Weeks 4 through 8, hold steady. Resist adding anything else. Most of these need six to eight weeks to show a real signal, and week-to-week fibromyalgia variability will fool you if you judge at day ten.

Week 9, review honestly. Compare your last two weeks of scores with your baseline two weeks. If nothing moved and nothing improved in how you feel, stop that product. Not every trial succeeds, and stopping is a result, not a failure.

Weeks 10 through 12, add the second thing. Same rules. One change, low end of the range, keep scoring.

Bring the numbers to your appointments. Twelve weeks of simple daily scores is more useful to your clinician than any description you can give from memory.

Realistic expectations

Let me be direct about what good looks like, because unrealistic expectations are how people quit three weeks before something would have helped.

What is realistic. Modest, real, cumulative gains. A point off average pain. Falling asleep faster. Two fewer flare days in a month. Getting through an afternoon without the crash. Those are wins that compound, and they are the kind of change the trials in this guide actually measured.

What is not realistic. A supplement making fibromyalgia go away. Anyone selling you that is selling you something. The studies behind even the better-supported ingredients here are mostly small, mostly short, and report improvements in scores rather than resolution of the condition.

Timeline. Most of these trials ran six to sixteen weeks. Give things a fair window before you judge, and do not judge on a flare week.

Combination effects. The most encouraging fibromyalgia supplement study in this guide tested supplements added on top of existing prescription treatment and structured care (PMID 37378482). That is the model I would point you toward: supplements as an addition to the foundation, not a substitute for it.

Cost sense. If a product has produced nothing after eight to twelve weeks at a reasonable dose, stop paying for it. That money is better spent on food you will actually cook, or on care that has evidence behind it.

When to seek professional care

Book with a clinician soon if:

- Your symptoms are getting worse rather than plateauing.
- Fatigue is severe, new, or dramatically different from your usual pattern.
- You have never had a proper diagnostic workup. Thyroid disease, anemia, sleep apnea, inflammatory arthritis, and other conditions can mimic or accompany fibromyalgia, and they are treated differently.
- You are on four or more prescription medications and are adding supplements.
- Sleep is broken despite good sleep habits, or your partner reports snoring or pauses in breathing.

Seek care urgently for any of these, because they are not fibromyalgia:

- New joint swelling, redness, or warmth, or a fever with your pain
- Unexplained weight loss
- New weakness, numbness, or loss of bladder or bowel control
- Severe headache unlike your usual pattern, or a sudden change in vision
- Chest pain or new shortness of breath
- Thoughts of harming yourself. Please call or text 988 in the United States for the Suicide and Crisis Lifeline, or go to an emergency department.

Signs to stop a supplement and call your prescriber right away: agitation, confusion, rapid heartbeat, heavy sweating, shivering, muscle twitching or rigidity, or diarrhea after starting a serotonin-active product. Those are the warning signs of serotonin syndrome (PMID 37309284). Unusual bruising or bleeding, yellowing of the skin or eyes, dark urine, a rash, or any new swelling also warrant a call.

Where to find quality versions of these

Nothing in this guide names a brand, and that is on purpose. The evidence supports ingredients, not labels. But supplement quality genuinely varies, so here is what to look for on your own.

Look for third-party testing. NSF Certified for Sport, USP Verified, or an equivalent independent seal means someone outside the company confirmed the bottle contains what the label claims, at the amount claimed, without common contaminants. This matters more in supplements than in almost any other consumer category, because the manufacturer, not a regulator, is responsible for verifying content before it goes on sale.

Check the form, not just the name. Magnesium glycinate and magnesium oxide are both “magnesium” on the front of the bottle and behave very differently. The forms this guide names in each entry are the forms studied.

Read the whole ingredient panel. Combination products for sleep, mood, or pain frequently include serotonin-active ingredients that do not appear on the front label. If you take any serotonin-active medication, this is where problems start.

If you would rather not evaluate all of that yourself, I keep a screened dispensary through Fullscript, which ships directly to you. I hold no inventory myself. Everything in it is screened against the quality standards above and organized by the categories in this guide.

> ***Root Health dispensary: us.fullscript.com/welcome/drseay***
>
> *Disclosure: Root Health earns a portion of dispensary purchases. That revenue supports the practice, and it is also why this guide was written to be complete and usable on its own. Every recommendation here is based on the published evidence cited in these pages, and nothing was included, excluded, ranked, or dosed based on what it earns. You can buy any of these anywhere, and several of the most useful items in this guide, including the sleep and movement work, cost nothing at all.*

Working with Root Health

Everything in this guide is general education, deliberately written to be useful without knowing your labs. A personalized plan is a different thing: it starts with testing, your medication list, your history, and the symptoms costing you the most, and it sequences interventions instead of stacking them.

If you want that, a Root Discovery Session is the right first step:

> ***Book a free Root Discovery Session: rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=fibromyalgia-supplement***

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

> ***Get in touch: rootcauseunlocked.com/contact?utm_source=guide&utm_medium=ebook&utm_campaign=fibromyalgia-supplement***

If you would rather start on your own, the most useful next step is to start tracking. Symptom patterns are hard to see from inside them, and a written record turns "I think the magnesium helped" into something you and any clinician can actually read:

> ***Free symptom tracker: rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=fibromyalgia-supplement***

Read the companion document, the **Fibromyalgia Diet Guide**, alongside this one. Several of the nutrients here work better inside an eating pattern that supports them, and the diet evidence is worth understanding on its own terms.

Reach me at (919) 662-0520 or at 503 US-70 East, Suite L, Garner, NC 27529.

References

All references below were verified against live PubMed records on 2026-08-02, including a check of publication type for retractions. Full verification detail, including what each citation does and does not support, is documented in `Fibromyalgia-Citation-Ledger-Supplements-Diet.md`.

1. Macfarlane GJ, Kronisch C, Dean LE, et al. EULAR revised recommendations for the management of fibromyalgia. *Ann Rheum Dis.* 2017;76(2):318-328. PMID 27377815
2. Jones EA, Asaad F, Patel N, et al. Management of Fibromyalgia: An Update. *Biomedicines.* 2024;12(6). PMID 38927473
3. Ali OME. Prevalence of Vitamin D Deficiency and Its Relationship with Clinical Outcomes in Patients with Fibromyalgia: a Systematic Review of the Literature. *SN Compr Clin Med.* 2022;4(1):38. PMID 35071984
4. Wepner F, Scheuer R, Schuetz-Wieser B, et al. Effects of vitamin D on patients with fibromyalgia syndrome: a randomized placebo-controlled trial. *Pain.* 2014;155(2):261-268. PMID 24438771
5. Qu K, Li MX, Zhou YL, et al. The efficacy of vitamin D in treatment of fibromyalgia: a meta-analysis of randomized controlled studies and systematic review. *Expert Rev Clin Pharmacol.* 2022;15(4):433-442. PMID 35596576
6. Bagis S, Karabiber M, As I, et al. Is magnesium citrate treatment effective on pain, clinical parameters and functional status in patients with fibromyalgia? *Rheumatol Int.* 2013;33(1):167-172. PMID 22271372
7. Boulis M, Boulis M, Clauw D. Magnesium and Fibromyalgia: A Literature Review. *J Prim Care Community Health.* 2021;12:21501327211038433. PMID 34392734
8. Ferreira I, Ortigoza A, Moore P. Magnesium and malic acid supplement for fibromyalgia. *Medwave.* 2019;19(4):e7633. PMID 31150373
9. Fattah MAA, Morsi S, Fattah SA, et al. High-dose Omega-3 Alters Serum Magnesium and Calcium Levels and Affects Fibromyalgia Symptoms: A Randomized, Double-blind, Placebo-Control Study. *Curr Rheumatol Rev.* 2025;21(4):446-454. PMID 39377412
10. Rondanelli M, Faliva MA, Miccono A, et al. Food pyramid for subjects with chronic pain: foods and dietary constituents as anti-inflammatory and antioxidant agents. *Nutr Res Rev.* 2018;31(1):131-151. PMID 29679994
11. Hussain SA, Al-Khalifa II, Jasim NA, Gorial FI. Adjuvant use of melatonin for treatment of fibromyalgia. *J Pineal Res.* 2011;50(3):267-271. PMID 21158908
12. Hemati K, Amini Kadijani A, Sayehmiri F, et al. Melatonin in the treatment of fibromyalgia symptoms: A systematic review. *Complement Ther Clin Pract.* 2020;38:101072. PMID 31783341
13. Salaffi F, Farah S, Sarzi-Puttini P, Di Carlo M. Palmitoylethanolamide and acetyl-L-carnitine act synergistically with duloxetine and pregabalin in fibromyalgia: results of a randomised controlled study. *Clin Exp Rheumatol.* 2023;41(6):1323-1331. PMID 37378482
14. Rossini M, Di Munno O, Valentini G, et al. Double-blind, multicenter trial comparing acetyl l-carnitine with placebo in the treatment of fibromyalgia patients. *Clin Exp Rheumatol.* 2007;25(2):182-188. PMID 17543140
15. Cordero MD, Alcocer-Gomez E, de Miguel M, et al. Can coenzyme Q10 improve clinical and molecular parameters in fibromyalgia? *Antioxid Redox Signal.* 2013;19(12):1356-1361. PMID 23458405
16. Rossello Aubach L, Fornes Roca X, Fernandez Alvarez ME. Effects of Coenzyme Q10, Tryptophan, and Magnesium Supplementation on Fatigue in Patients with Fibromyalgia: A Randomized Trial. *J Diet Suppl.* 2025;22(3):433-444. PMID 40151031

17. Alves CR, Santiago BM, Lima FR, et al. Creatine supplementation in fibromyalgia: a randomized, double-blind, placebo-controlled trial. *Arthritis Care Res (Hoboken).* 2013;65(9):1449-1459. PMID 23554283

18. Minerbi A, Gonzalez E, Brereton NJB, et al. Altered microbiome composition in individuals with fibromyalgia. *Pain.* 2019;160(11):2589-2602. PMID 31219947

19. Aslan Cin NN, Acik M, Tertemiz OF, et al. Effect of prebiotic and probiotic supplementation on reduced pain in patients with fibromyalgia syndrome: a double-blind, placebo-controlled randomized clinical trial. *Psychol Health Med.* 2024;29(3):528-541. PMID 37224267

20. Roman P, Estevez AF, Miras A, et al. A Pilot Randomized Controlled Trial to Explore Cognitive and Emotional Effects of Probiotics in Fibromyalgia. *Sci Rep.* 2018;8(1):10965. PMID 30026567

21. Calandre EP, Hidalgo-Tallon J, Molina-Barea R, et al. The Probiotic VSL#3 Does Not Seem to Be Efficacious for the Treatment of Gastrointestinal Symptomatology of Patients with Fibromyalgia: A Randomized, Double-Blind, Placebo-Controlled Clinical Trial. *Pharmaceuticals (Basel).* 2021;14(10):1063. PMID 34681287

22. Caruso I, Sarzi Puttini P, Cazzola M, Azzolini V. Double-blind study of 5-hydroxytryptophan versus placebo in the treatment of primary fibromyalgia syndrome. *J Int Med Res.* 1990;18(3):201-209. PMID 2193835

23. Mikkelsen N, Damkier P, Pedersen SA. Serotonin syndrome: A focused review. *Basic Clin Pharmacol Toxicol.* 2023;133(2):124-129. PMID 37309284

24. Jacobsen S, Danneskiold-Samsoe B, Andersen RB. Oral S-adenosylmethionine in primary fibromyalgia. Double-blind clinical evaluation. *Scand J Rheumatol.* 1991;20(4):294-302. PMID 1925418

25. Tavoni A, Jeracitano G, Cirigliano G. Evaluation of S-adenosylmethionine in secondary fibromyalgia: a double-blind study. *Clin Exp Rheumatol.* 1998;16(1):106-107. PMID 9543578

26. Nicolussi S, Drewe J, Butterweck V, Meyer zu Schwabedissen HE. Clinical relevance of St. John's wort drug interactions revisited. *Br J Pharmacol.* 2020;177(6):1212-1226. PMID 31742659

27. Boomershine CS, Koch TA, Morris D. A Blinded, Randomized, Placebo-Controlled Study to Investigate the Efficacy and Safety of Ferric Carboxymaltose in Iron-Deficient Patients with Fibromyalgia. *Rheumatol Ther.* 2018;5(1):271-281. PMID 29149437

28. Lundh A, Lexchin J, Mintzes B, et al. Industry sponsorship and research outcome. *Cochrane Database Syst Rev.* 2017;2(2):MR000033. PMID 28207928

Dr. Adam Seay, DC · Root Health · Garner, NC · rootcauseunlocked.com